

Bachelor's Thesis
**KAIZEN STRATEGIES FOR IMPROVING PRODUCTION FLOW IN
PHARMACEUTICAL ENVIRONMENTS**

Double degree in Chemical Engineering and Industrial Electronics and Automation.

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Report and Annex

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Abstract

This Bachelor's Final Thesis reports on a curricular internship conducted at Kaizen Institute in collaboration with Sincrofarm, a medium-scale pharmaceutical manufacturer, and situates the work at the application of academic Lean-manufacturing to production optimization. The project's explicit aim was to analyze Sincrofarm's primary bottling and packaging lines, identify improvement opportunities, and implement Kaizen guided interventions to reduce operational costs and increase throughput while preserving product quality and regulatory compliance.

A quantitative data driven methodology was applied: on-site Gemba observations, time-studies, and baseline data collection were combined with Lean and Kaizen tools; SMED for setup reduction, Line Balancing and Yamazumi charts for workload redistribution, coupled with standard work and visual management for sustained standardization. The approach emphasized elimination of non-value-added activities, resource optimization, and iterative improvements validated through responsive data analysis.

Interventions produced measurable performance gains: demonstrated increases in Overall Equipment Effectiveness (OEE) and line productivity metrics, a calculated annual cost reduction of 112,000€, and a projected return on investment within three months of full implementation. Improvements were achieved primarily through reduced changeover times, improved labor allocation, decreased idle time, and more consistent cycle times.

Beyond these quantitative outcomes, the project yielded broader strategic benefits: strengthened operator engagement through participatory Kaizen activities, enhanced process robustness supportive of regulatory requirements, and reduced material and energy waste contributing to ecological sustainability. These results illustrate that disciplined application of Lean in a pharmaceutical setting can deliver rapid financial returns, empower the workforce, and support longer-term competitiveness and sustainability. The thesis concludes with recommendations for scaling the methodology across additional product families and for integrating continuous improvement into Sincrofarm's operational governance.

Resum

Aquest Treball de Fi de Grau informa sobre unes pràctiques curriculars realitzades amb la consultora Kaizen Institute en col·laboració amb Sincrofarm, a la seva planta de fabricació farmacèutica, i situa l'obra en l'aplicació del Lean manufacturing acadèmic a l'optimització de la producció. L'objectiu explícit del projecte és analitzar les principals línies d'embotellat i d'envasat de Sincrofarm, identificar oportunitats de millora i implementar intervencions amb mètodes Kaizen per reduir els costos operatius i augmentar el rendiment, tot preservant la qualitat del producte i el compliment de la legislació.

S'aplica una metodologia quantitativa basada en dades empíriques: observacions en el *Gemba* (*in situ*), estudis de temps i recollida de dades de referència. Combinat amb eines Lean i Kaizen com; l'SMED per a la reducció dels temps de preparació, equilibrat de línies i gràfics Yamazumi per a la redistribució de la càrrega de operativa, juntament amb el treball estàndard i la gestió visual per aconseguir una estandardització sostinguda. L'enfocament del treball fa èmfasi en l'eliminació d'activitats de no valor afegit, l'optimització dels recursos i les millores iteratives validades mitjançant una anàlisi de dades de resposta ràpida.

Les intervencions produeixen guanys en el rendiment global de la planta mesurables: increments demostrats en l'Eficàcia Global dels Equips (OEE) i en els indicadors de productivitat de les línies, una reducció anual de costos calculada de 112.000€, i un retorn de la inversió de tres mesos. Les millores s'aconsegueixen principalment mitjançant la reducció dels temps de canvi, una millor assignació de la mà d'obra, la disminució dels temps d'inactivitat i velocitats més consistents.

Més enllà d'aquests resultats quantitius, el projecte aporta beneficis estratègics més amplis: major compromís dels operaris a través d'activitats participatives de Kaizen, robustesa de processos reforçada i alineada amb els requisits legals, reducció del malbaratament de materials i energia que contribueix a la sostenibilitat ecològica. Aquests resultats il·lustren que l'aplicació disciplinada del Lean i el Kaizen en un entorn farmacèutic pot oferir rendiments financers ràpids, empoderar la plantilla i generar competitivitat i sostenibilitat a llarg termini. El treball conclou amb recomanacions per escalar la metodologia a altres famílies de productes i per integrar la millora contínua dins la governança operativa de Sincrofarm.

Resumen

Este Trabajo de Fin de Grado presenta el trabajo realizado bajo las prácticas curriculares realizadas en Kaizen Institute en colaboración con Sincrofarm, un fabricante farmacéutico de mediana escala, y sitúa el trabajo en la aplicación del Lean manufacturing académico a la optimización de la producción. El objetivo explícito del proyecto fue analizar las principales líneas de embotellado y envasado de Sincrofarm, identificar oportunidades de mejora e implementar intervenciones guiadas por Kaizen para reducir los costes operativos y aumentar la productividad, preservando al mismo tiempo la calidad del producto y el cumplimiento legislativo.

Se aplicó una metodología cuantitativa basada en datos: observaciones en el *Gemba*, estudios de tiempos y recogida de datos de referencia, combinados con herramientas de Lean y Kaizen; SMED para la reducción de tiempos de cambio, Equilibrado de Líneas y gráficos Yamazumi para la redistribución de la carga operacional, junto con trabajo estandarizado y gestión visual para una estandarización sostenida. El enfoque puso énfasis en la eliminación de actividades de no valor añadido, la optimización de recursos y la mejora iterativa validada mediante análisis de datos en tiempo real.

Las intervenciones produjeron mejoras de rendimiento medibles: incrementos demostrados en la Eficiencia Global de los Equipos (OEE) y en los indicadores de productividad de las líneas, una reducción de costes anual calculada en 112.000€, y un retorno de la inversión de tres meses. Las mejoras se lograron principalmente mediante la reducción de los tiempos de cambio, una mejor asignación de la mano de obra, la disminución del tiempo improductivo y una mayor consistencia en las velocidades de trabajo.

Más allá de estos resultados cuantitativos, el proyecto generó beneficios estratégicos más amplios: fortalecimiento del compromiso de los operarios a través de actividades Kaizen participativas, mayor robustez de los procesos en apoyo de los requisitos regulatorios, y reducción de residuos materiales y energéticos que contribuyen a la sostenibilidad ecológica.

Estos resultados ilustran que la aplicación disciplinada de Lean y Kaizen en un entorno farmacéutico puede generar retornos financieros rápidos, empoderar a la plantilla y respaldar la competitividad y sostenibilidad a largo plazo. El trabajo concluye con recomendaciones para escalar la metodología a otras líneas y para integrar la mejora continua en la gestión operativa de la empresa.

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Glossary

Glossary

Eighty-twenty Rule Refers to the Pareto principle, which states that roughly 80% of effects come from 20% of the causes. [15](#)

Gemba A Japanese word meaning "the real place." In a Kaizen context, it usually refers to the place where value is added, such as the shop floor. It can also refer to any place where work is being performed. [9](#)

GQCDM Principles An acronym for Growth, Quality, Cost, Delivery, and Motivation. It represents a strategic framework used in business and manufacturing to guide performance improvements.. [12](#), [19](#)

JIT Just In Time. A system that "pulls" parts through production based on customer demand rather than "pushing" them based on projected demand. It results in line balancing and little to no excess inventory. [8](#), [9](#)

Kaizen A Japanese term meaning "change for the better." It is a gradual, long-term approach that focuses on small, incremental changes to improve efficiency and quality. It was popularized by Masaaki Imai. [6](#), [8–10](#), [12](#), [13](#), [18](#), [19](#), [22](#), [26](#)

KPI An English acronym for Key Performance Indicators, which are the key indicators of a given process or activity. [10](#), [11](#), [13](#), [23](#), [26](#)

Lean An English term meaning "no fat." In a management context, it is a strategy to reduce or eliminate waste or excess. [8](#), [9](#)

Muda A Japanese word for "Waste" and a key concept in TPS. It is one of three types of deviation from optimal resource allocation (Muda, Mura, and Muri). [8](#), [9](#), [13](#)

OPL Stands for One Page Lesson. Is a simple, visual, and concise training document used on the shop floor to quickly and effectively transfer knowledge. The key characteristic of an OPL is its brevity; it contains all the necessary information for a specific task or procedure on a single page. OPLs are typically created and maintained by the operators themselves, ensuring the content is practical and relevant. They are a core tool in Lean Manufacturing for standardizing best practices, providing quick reference for machine setup or troubleshooting, and promoting continuous improvement. They serve as a powerful alternative to lengthy manuals, making training accessible and immediate.. [16](#)

SMED Stands for **Single-Minute Exchange of Die**. A Lean manufacturing methodology for drastically reducing the time it takes to complete equipment changeovers. The goal is to make changeovers so quick that it becomes economically viable to produce smaller batch sizes, reducing inventory and improving production flexibility.. [15](#), [16](#), [18](#)

TPS Toyota Production System. A methodology that resulted from over 50 years of Kaizen at Toyota. It is built on a foundation of Leveling with supporting pillars of Just-in-Time and Jidoka. [8](#)

Visual Management A set of techniques for creating a workplace that uses visual communication and control. The philosophy is "what gets measured and displayed, gets done." Simple tools are used to identify the target state and correct any deviations. [9](#)

VSA Stands for Value Stream Analysis. A Lean management method used to visually map and analyze the flow of materials and information required to deliver a product or service to a customer. VSA identifies value-added and non-value-added activities, highlights waste, and provides a basis for Kaizen (continuous improvement) efforts. It typically includes a current-state map, future-state vision, and an actionable plan to close gaps.. [14](#), [18](#)

Preface

Project Origin

This project is part of the curricular internship carried out at the consulting firm Kaizen Institute, which began in February 2025. Kaizen Institute specializes in continuous improvement, a concept that will be defined in the first chapter (1) as it provides the framework for understanding this project's approach.

The project originated when *Sincrofarm*, a pharmaceutical manufacturing company, decided to initiate a transformation process in its facility and contacted Kaizen Institute for guidance and advisory support throughout the process.

The company's production model is similar to that of a laboratory: it operates in sections, producing different elements separately. Each section has specific cycle times and production capacities that are not always compatible with those of other sections, resulting in waiting times, overproduction, or, in some cases, underproduction. Furthermore, due to stringent internal quality standards and regulatory requirements, a significant portion of operating time is spent on non-value-added activities, such as cleaning equipment or performing machine changeovers. This way of working has been effective for over forty years, but the company's exponential growth now prevents it from meeting order demand consistently, limiting its competitiveness in the sector. For this reason, modernization and adaptation to customer demand are essential to meet deadlines and increase profitability.

The project discussed below is extensive and divided into two stages: increasing line efficiency and enhancing line productivity. A daily management system called Daily Kaizen was also implemented during the course of the project; however, the details of this system are beyond the scope of the present work.

Motivation

This project is highly rewarding both personally and professionally, as it presents the challenge of completely transforming a production plant, overseeing the proposed changes, and supporting the personnel throughout the process. It provides the opportunity to drive a significant positive impact within the company, not only economically but also in terms of the work environment, by considerably improving both working conditions and ways of working.

One of the personal motivations behind this project is witnessing the practical application of my work and seeing that it has a tangible purpose. Furthermore, the project is complex: it required me to apply the knowledge gained during my university studies while also challenging my perspective, as I often had to address both technical and managerial aspects, the latter being a new experience for me. Many changes needed to be implemented, which inevitably led to unforeseen challenges along the way. This process allowed me to grow professionally, learning to tackle problems proactively, find solutions, and adapt without letting obstacles take control. Growth comes from stepping outside our comfort zone, and this project has contributed significantly to my development as a professional.

The greatest motivation, however, comes from seeing tangible results: observing the project from start to finish, with all its unforeseen challenges, is immensely rewarding. It provides a complete perspective of the process and enables the extraction of lessons and improvements not only for this project but also for future initiatives.

Planning and Project Stages

The project was delivered in the following stages:

Task / Phase	Duration (days)	Start Date	Finish Date
Study of Kaizen Production System	21	10/01/2025	07/02/2025
Initial Planning and Data Extraction Methodology			
Initial Planning	1	10/02/2025	10/02/2025
Factory Visit	1	17/02/2025	17/02/2025
Creation of Data Templates	3	19/02/2025	21/02/2025
Current State Analysis			
Data Recording Start	1	24/02/2025	24/02/2025
Shadowing (Jars / Liquids)	2	25/02/2025	26/02/2025
Solution Design			
Kaizen Boards	1	27/02/2025	27/02/2025
User Training	1	05/03/2025	05/03/2025
SMED Workshop	3	03/03/2025	05/03/2025
Implementation			
Daily Kaizen Kick-off	1	03/03/2025	03/03/2025
Cleaning Standard Procedure	2	06/03/2025	07/03/2025
General Standard of the Worker	2	14/03/2025	20/04/2025
Rollout			
Validation	31	10/03/2025	31/07/2025
Data Analysis	12	10/03/2025	31/07/2025
Control			
Steering Committee	6	17/02/2025	31/07/2025

Table 1: Summary of project tasks and durations by phase

In the Appendix A, Figure 27 a much more detailed Gantt Chart of the project phases is included for further reference.

1 Introduction

1.1 Production Management

1.1.1 Traditional Production Management

This project takes place under the discipline of Production Management.

Production management is the discipline concerned with planning, organizing, directing, and controlling the production processes of goods and services. Its central goal is to transform inputs, such as raw materials, labor, and technology. Into outputs in a efficient, effective, and sustainable manner. In essence, it ensures that products are made at the right quality, in the right quantity, at the right time, and at the right cost.

As explained by Womack at *The Machine that changed the world* [1], the history of production management can be traced back to ancient civilizations, where organized methods were developed for building monuments, managing agriculture, and crafting tools. The craft production system dominated for centuries, where skilled artisans produced customized goods by hand. But the Industrial Revolution (18th–19th century) marked a turning point [2]. The introduction of mechanization, steam engines, and factories shifted production from small workshops to mass production systems.

Division of labor, popularized by Adam Smith, increased productivity by assigning specific tasks to workers. Frederick W. Taylor introduced scientific management, emphasizing time-and-motion studies, standardization, and efficiency [3].

Henry Ford pioneered the assembly line system in the early 20th century, drastically reducing production times and costs. While these systems brought unprecedented efficiency, they were also characterized by rigidity, limited flexibility, and an overemphasis on output quantity [2]. By the mid-20th century, it became evident that sheer productivity was insufficient to sustain competitiveness, especially as consumers began to demand higher quality, reliability, and variety in products [4].

1.1.2 Post–World War II Developments

After World War II in Japan’s Toyota Production System (TPS) was introduced giving birth to Lean manufacturing. It was when Ohno published his main work *Toyota Production System* focusing on waste reduction, continuous improvement (**Kaizen**), and just-in-time (JIT) production.

These practices reshaped global manufacturing by prioritizing quality and efficiency [1].

By the late 20th century, the competitive edge of organizations was no longer determined by cost leadership alone. Globalization and increasingly demanding markets required production systems capable of delivering consistent quality, adaptability, and customer responsiveness. This led to the widespread adoption of approaches such as Total Quality Management (TQM), Six Sigma, and Lean thinking, all of which placed the pursuit of continuous improvement and error prevention at the core of industrial practice [1].

Production management has expanded beyond manufacturing to include service industries, project-based industries, and digital production.

The rise of Industry 4.0 (automation, IoT, artificial intelligence, and smart factories) continues to redefine the field, emphasizing flexibility, customization, and data-driven decision-making [5].

An equally important shift is occurring in the role of the workforce. Rather than viewing employees as replaceable operators of machinery, emerging production philosophies recognized their capacity to

identify inefficiencies, propose improvements, and contribute to organizational learning. This cultural shift laid the groundwork for systems that integrate human creativity with structured methods for ongoing enhancement.

The following table 1 summarizes the difference between Lean Manufacturing and the Traditional Bulk Manufacturing Philosophy.

Feature	Lean Manufacturing	Traditional Marketing
Production Model	Pull System – Production based on real-time demand	Push System – Production based on forecasts
Waste Management	Eliminates waste through Continuous Improvement	High waste due to excess inventory and overproduction
Inventory Management	Just-in-Time (JIT) – Minimum stock levels maintained	Large inventories held to prevent shortages
Flexibility	High – Easily adapts to demand changes	Low – Fixed processes make rapid adjustments difficult
Employee Involvement	High – Encourages problem-solving and efficiency improvements	Low – Employees follow standardized tasks with limited decision-making
Process Optimization	Uses Value Stream Mapping and Standardized Work to refine processes	Rigid processes with minimal optimization

Figure 1: Key differences between Lean Manufacturing vs Traditional Production Management. Source. [6]

1.1.3 The KAIZEN Methodology

After World War II, Toyoda Kiichiro, then president of the Toyota Motor Company, saw the need to catch up with America to assure the survival of the automobile industry of Japan. Taiichi Ohno, Toyota's engineer, had a challenge: building a diversified line of products with the limited equipment Toyota possessed at the time [2]. Instead of using batch production, he designed a system of integrated production that used one-piece flow, demonstrating that a trade-off between quality and productivity does not need to exist [7]. Aligned with other practices, Toyota Production System (TPS) was born to defy the premise that bigger batches mean lower costs.

TPS, which ultimate goal is the absolute elimination of waste, is based in two pillars: just-in-time (JIT) and autonomation (automation with a human touch)[2]. The lack of the necessary capital resources to support high inventory levels was solved with the implementation of a just-in-time pull system. JIT means "to produce the necessary units at the necessary quantities at the necessary time". A company establishing this flow throughout can approach zero inventory. With autonomation, Toyota intended to create a built-in automatic checking system against small abnormalities [8]. With this prevention mechanism, defective products were not produced. That was not the only advantage. While the machine is working properly, employees are liberated. Consequently, one worker could attend multiple machines, increasing production efficiency. Furthermore, autonomation came as a facilitator of improvement. Stopping the machine creates awareness and, through the total comprehension of the problem, improvement measures can be taken [2]. Lean, based on TPS, is a methodology that accentuates customer needs, to improve quality and reduce costs through continuous improvement and employee involvement [9]. Kaizen, a Japanese word, refers to the concept of continuous improvement. It does not intend to be mistaken as innovation or disruption and elevated costs. On the other hand, Kaizen is about methodological improvements. It is a low-cost low-risk process that assures incremental progress and sustainable changes in the long term.

At the beginning of the 21st century, Toyota Motor Company surpassed General Motors and became the world leader in automobile production, expanding and elevating the Kaizen concept for its key role in the company's success [7].

Kaizen's methodology is based on five fundamental principles described in Massaki and Sanchez's

book [4] :

- Create customer value

Value is what the customer is willing to pay for. Using a market-in approach to make informed decisions based on what the customers want and offer them that in the best way possible, improving customer experience.

- Create flow efficiency

Flow efficiency can be obtained through the elimination of three Ms: **Muda** (waste), **Mura** (unevenness) and **Muri**(overburden). The concept of **Muda** primarily originated from Taiichi Ohno's production philosophy in the early 1950s. He defined, in the industrial context he was inserted, seven sources of **Muda**: overproduction, waiting, inventory, motion, transportation, over-processing and defects. All the activities that do not add value to the process are considered **Muda**. Fujio Cho, former President of Toyota, defined waste as "anything other than the minimum amount of equipment, materials, parts, space and worker's time, which are absolutely essential to add value to the product". Eliminating waste is the most effective way to increase productivity and decrease costs. Although **Muda** has gained bigger awareness, comprehending all three concepts allows for a greater understanding of **Lean** [7]. **Mura** is the variation in a process that is not caused by the final customer, for example highs and lows in the planned production due to the production system or the irregular work rhythm. Leveling production using **JIT** can eliminate this irregularities, avoiding work spikes and long waiting times for workers. **Muri** means to overburden equipment or workers, demanding a faster rhythm for a longer period than expected. While in equipment it can provoke defects and failures, in people it can result in safety issues.

- Be **Gemba** oriented

Gemba is the place where value is added. Problems should be identified and solved there, at their root cause. Central to Kaizen is *genchi genbutsu* (go and see for yourself). Going to the **Gemba** means going to the place where the action really happens, to be able to thoroughly observe the reality of the processes.

- Empower people

Respect for people is a fundamental principle of the philosophy. It recognizes that people are not the problem, processes are. It intends to improve every day, everywhere, with everyone, training all employees and encouraging people to think and share their own improvement ideas, systematically. It is in management's biggest interest to create the best possible conditions for everyone to do the best possible work.

- Be scientific and transparent

As important as identifying problems, is knowing how to explain and substantiate one's findings, supporting them with data. **Visual Management** is a basal tool of Lean Methodology, that highlights problems and allows for a quicker reaction to them.

The methodology is supported by a number of tools and systems. Some of which are used in this work, like the **SMED** or the Yamazumi.

1.1.4 Brief introduction to the SMED

The **SMED** is a technique that helps to significantly reduce the time required for machine changeovers and setups. Developed by Shigeo Shingo in the 1950s [10], the core principle is to convert as many setup operations as possible from *internal* tasks (those that can only be performed when the machine is stopped) to *external* tasks (those that can be completed while the machine is still running). The ultimate goal is to complete changeovers in less than ten minutes (a "single minute" in the literal sense of the word), thereby maximizing production time and enabling smaller, more flexible batch sizes.

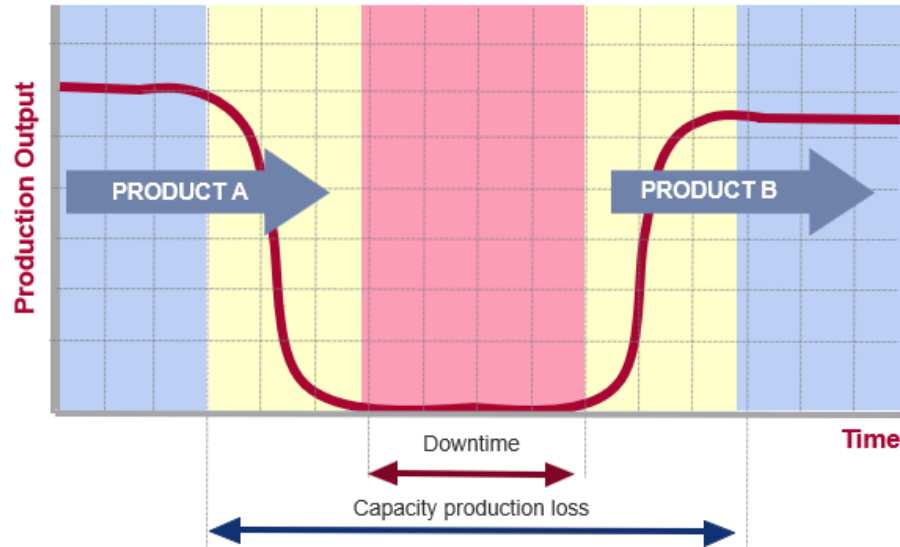


Figure 2: Down time visualization. The arrows represent the production material flow. The red line represents production output rate at specific times. The blue, yellow and red zones represents the nominal production output, reduced production output rate, and no production output, respectively. Source: Internal Kaizen Institute documents.

As detailed in Shingo's foundational book, *A Revolution in Manufacturing* [10], this methodology is not limited to product changeovers but can be applied to any process with internal and external tasks, such as the cleaning procedure identified. The SMED implementation process follows these distinct steps:

1. **Preliminary Stage:** Identify and analyze the current setup or cleaning process. This involves documenting all tasks and their timings.
2. **Stage 1: Separate Internal and External Tasks:** Critically evaluate each step of the process and categorize it as either internal or external.
3. **Stage 2: Convert Internal from External Tasks:** This is the most crucial phase. Brainstorm and implement changes that allow internal tasks to be performed while the machine is running. For example, blowing off compressed air while the line is up.
4. **Stage 3: Streamline All Remaining Tasks:** Once the conversion is complete, focus on improving the efficiency of the remaining internal and external tasks. This involves using tools

like checklists, visual aids, and improved tool organization. For instance, using quick-release clamps instead of bolts.

5. **Sustain and Standardize:** Create and maintain standardized work instructions, checklists and OPLs to lock in the improvements. Establish a system for measuring and tracking progress to identify future opportunities.

The steps of the SMED methodology are depicted in Figure 3:

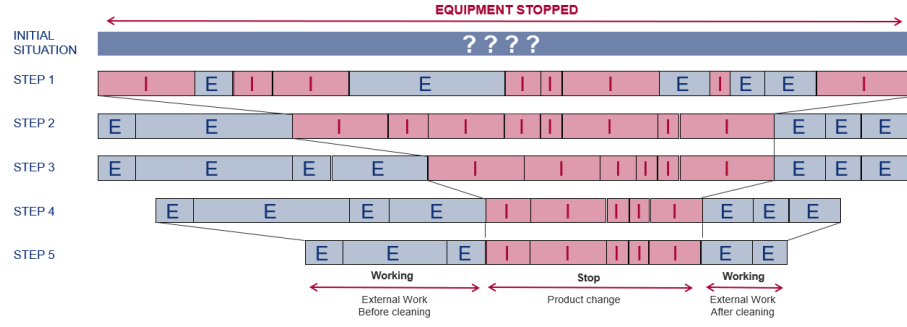


Figure 3: The evolution of the 5 SMED steps. The blue boxes marked with *E* represent *External* Tasks. The red boxes marked with *I* represent *Internal* tasks. Source: Internal Kaizen Institute Documentation

This structured approach ensures a systematic and measurable reduction in changeover or cleaning time, which directly increases equipment availability.

1.1.5 A Brief Description on Line Balancing and Yamazumi

The foundational principle of line balancing is the equal distribution of production tasks among operators. This approach aims to eliminate bottlenecks and minimize idle time, thereby optimizing workflow. The key analytical tool used for this work was the Yamazumi chart. This visual method, which literally translates to "to pile up," provided a graphical representation of each operator's work content and the time required for their individual tasks. By visualizing the cumulative workload for all team members, the Yamazumi chart immediately highlighted the imbalances and bottlenecks that were impeding the process. This visual data was critical for identifying and eliminating the forms of waste known as muda, mura, and muri—specifically, the waste of waiting, unevenness, and overburden—that were inherent in the unbalanced line. Through the strategic use of the Yamazumi chart, tasks were redistributed to create a smoother, more efficient, and balanced workflow, which directly contributed to the overall objective of maximizing productivity.

In conjunction with line balancing, the *Takt Time* is a critical metric for determining the required pace of production to meet customer demand. It serves as a heartbeat for the production process, establishing a baseline against which the balanced line's performance can be measured. The formula for calculating Takt Time is defined as (1):

$$\text{Takt Time} = \frac{\text{Total Available Production Time}}{\text{Customer Demand}} \quad (1)$$

The *Takt Time* will be used to determine the number of operators needed on the line in order to cover customer demand. As an example let's look onto the planned production for one shift. Let's

say it is 26,400 units that must be covered in one shift of 7.33h. In this case following equation 1, the *Takt Time* must be:

$$\text{Takt Time} = \frac{7,33h \cdot \frac{3,600s}{1h}}{26,400\text{units}} = 0.99\text{units/s} \quad (2)$$

Meaning that the line should be averaging a production of roughly 1 unit per second.

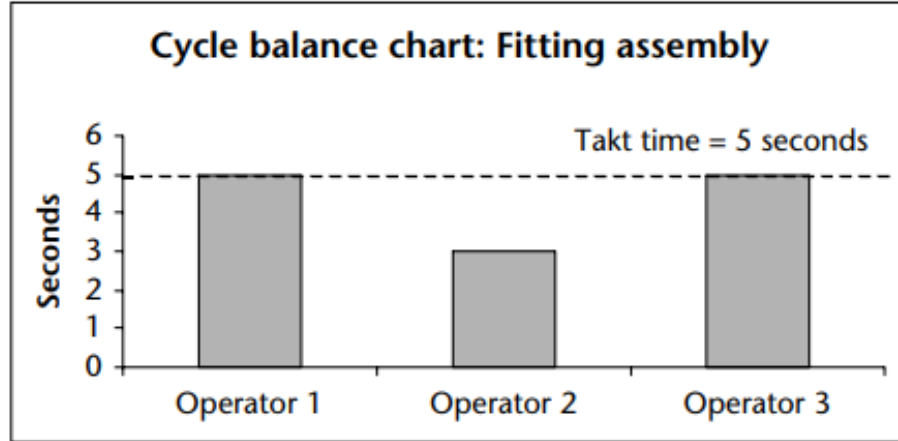


Figure 4: Yamazumi Chart Example. Source [8]

1.2 Plant and Problem Description

Sincrofarm is a pharmaceutical company located in Catalonia, specializing in Contract Manufacturing and Packaging Services for medicines and dietary supplements, both for human and veterinary use.

Its facilities are equipped with state-of-the-art machinery and cover a total surface area of 7,200 m², distributed across two specialized plants: one dedicated to pharmaceutical manufacturing and the other to nutraceuticals.

Some of the main products manufactured include:

- Collagen supplements
- Magnesium citrate supplements
- Lactase-based solutions
- *Lactobacillus*-based probiotics

The company employs more than 400 people, with approximately 80% of the workforce working directly on the shop floor.

The company encompasses various departments, each responsible of a part of the value stream. The focus of this work will be in the production lines.

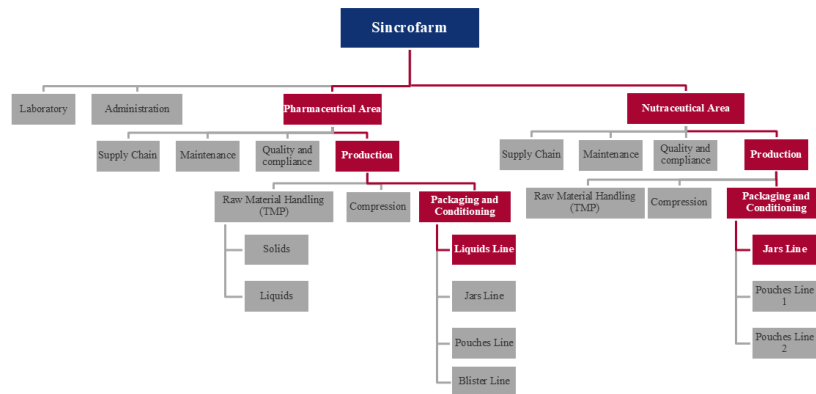


Figure 5: Sincrofarm's Departments and areas overview. Marked in red are the areas affected by the project scope.

Production is organized into several manufacturing areas and work lines:

- **TMP (Raw Materials Handling)**: Responsible for preparing and mixing the active components, whether liquids or solids, with the required excipients.
- **PI (Compression)**: Dedicated to compressing solid mixtures into tablets.
- **PAC (Packaging and Conditioning)**: Responsible for product presentation and packaging, with three main formats:
 - **Liquids**: Bottled liquid products.

- **Jars:** Solid tablets packaged in containers.
- **Pouches:** Single-use sachets containing uncompressed solids.

The material is transformed in every main area. The material flow is depicted in Figure 6

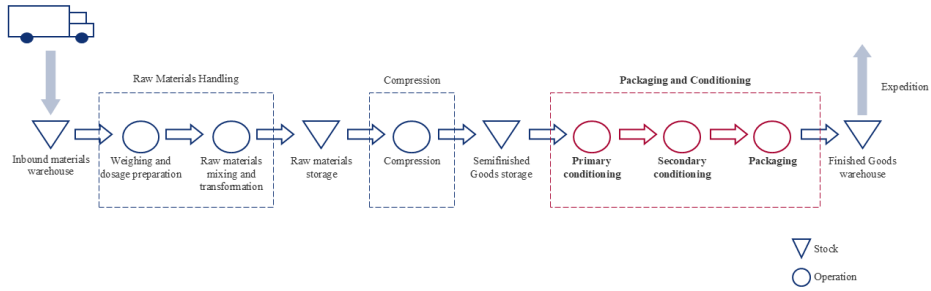


Figure 6: Sincrofarm's Material Flow chart. Marked in red are the areas affected by the project scope.

Each area works with a Semi-continuous production flow where long batches of a particular product are produced interrupted only by some planned and scheduled tasks like, cleaning, machine changeovers, etc.

In picture 7 a diagram of the Production Flow for each area is shown.

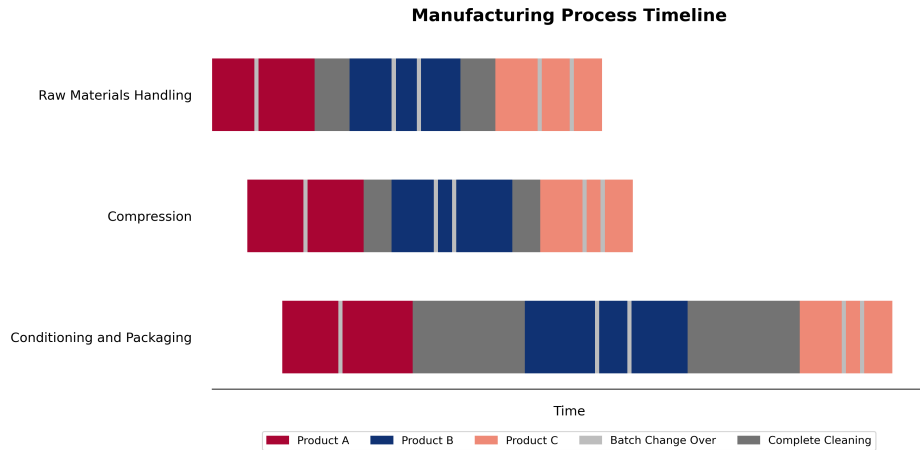


Figure 7: Sincrofarm's Production Flow Description

The focus of this work is on the gray-shaded activities shown in Figure 7. These activities occur during production time but do not contribute to transforming the product, and therefore are classified as non-value-added.

Accordingly, this project is dedicated to the elimination of non-value-added activities. Other factors that may influence production flow, such as line sequencing or planning, are considered outside

the scope of this study. Though the improvements achieved here are expected to have positive indirect effects on them as well.

1.3 Goals and pilots

The ultimate aim of this work is to deliver tangible value to Sincrofarm by ultimately reducing production costs by creating more efficient and standardized operations.

By focusing on specific, measurable aspects of the production process, the project seeks to identify bottlenecks, eliminate waste, and implement solutions that generate rapid financial and operational benefits.

To achieve this, two main subprojects were selected: **Improvement of Line Efficiency** and **Improvement of Line Productivity**. While related, these concepts address distinct dimensions of operational performance.

- Improvement of Line Efficiency.

Pilot: "Jars" Line

KPI: Line OEE

Goal: 40% OEE

- Improvement of line Productivity

Pilot: "Liquids" Line

KPI: Line Productivity

Goal: 12.5% $\Delta Productivity$

Shown in Figure 5 with red markings are the Pilots lines of this project.

2 Solution and Implementation

2.1 Improvement of Line Efficiency

2.1.1 Objective Function

As in any optimization problem, the objective function must be defined. To ensure alignment with the company strategy, improvement efforts are directed toward what is most valuable for the client, often summarized by the heuristics of [GQCDM Principles](#):

- **Growth:** Focus on expanding market share, innovation, and business development.
- **Quality:** Ensuring high standards in products, processes, and customer satisfaction.
- **Cost:** Controlling expenses and improving efficiency to remain competitive.
- **Delivery:** Meeting deadlines and ensuring timely, reliable delivery of products or services.
- **Motivation:** Encouraging and empowering employees to foster engagement, productivity, and innovation.

In this particular project, the focus will be on minimizing **Production Cost**. Specifically, this is pursued by maximizing the sustainable production capacity of the line.

$$\text{Objective function} = \max \left(\frac{\text{Total output units}}{\text{Time window}} \right) \quad (3)$$

In other words, equation 3 seeks to maximize the throughput rate, for example the number of orders the factory can process in a year.

This can also be expressed as:

$$\text{Objective function} = \max (\text{Rated line speed} \cdot OEE) \quad (4)$$

Here, the rated line speed represents the theoretical design capacity, while *OEE* adjusts for availability, performance, and quality losses. The introduction of this efficiency factor will be discussed in the following section.

It is evident that production output is strongly embedded in the cost structure of a company. A significant share of manufacturing cost is related to the utilization of machines and operators. By producing more in less time, the cost per unit decreases, thereby improving overall competitiveness.

2.1.2 Key Performance Indicators Selection

As a means to control how much the Production Capacity is affected by the results of this project. A [KPI](#) is chosen. For this particular project the Overall Equipment Efficiency (**OEE**) KPI is selected. Figure 8 the relationship between KPIs is visually depicted.

The OEE will impact directly to the production capacity via equation 4 and thus to the objective function.

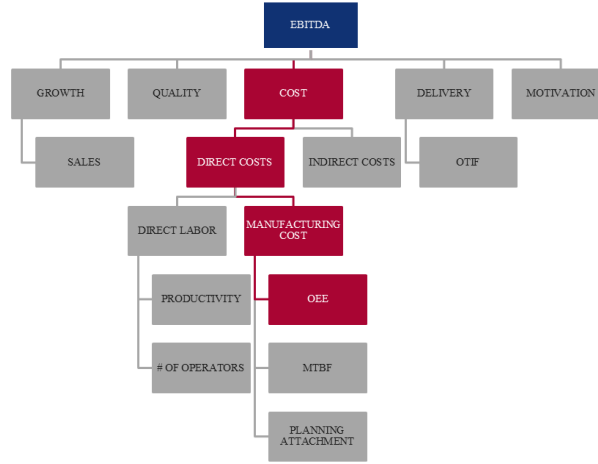


Figure 8: KPI Tree depicting relationship between indicators. Marked in red are the KPIs selected for the project.

2.1.2.1 OEE and profits calculation method

Overall Equipment Effectiveness (OEE) is a standard metric used to measure the efficiency of a production line. It identifies how much of the planned production time is truly productive. OEE is calculated by multiplying three core components: [Availability](#), [Performance](#), and [Quality](#).

2.1.2.2 Availability

Availability measures the proportion of scheduled production time during which equipment is operational and ready for use. It accounts for planned downtime, such as maintenance or shift breaks, as well as unplanned losses caused by equipment breakdowns or delays.[\[11\]](#)

$$A = \frac{\text{Operating Time (h)}}{\text{Planned Time (h)}} \quad (5)$$

With that definition, first 2 main terms may be acknowledged:

1. Planned Time: The total scheduled production time.
2. Operating time: The Planned Time minus all planned stoppages like changeovers, cleaning, and scheduled breaks.

2.1.2.3 Performance

Performance evaluates whether equipment is running at its optimal speed during production. It captures losses due to reduced speed or minor stoppages that may not halt production but reduce efficiency.[\[11\]](#)

$$\mu = \frac{\text{Actual Units Produced (units)}}{\text{Operating Time (h)} \cdot \text{Nominal Speed (units/h)}} \quad (6)$$

2.1.2.4 Quality

Quality assesses the proportion of defect-free products produced relative to the total output. This metric highlights losses due to rework or defective items, which reduce the overall effectiveness of the production process.[11]

$$Q = \frac{\text{Total OK unit output (units)}}{\text{Actual Units Produced (units)}} \quad (7)$$

2.1.2.5 OEE calculation

By combining these three components, the final OEE formula is derived, which shows the overall efficiency of the production line.

$$OEE = A \cdot \mu \cdot Q \quad (8)$$

An OEE score of 100% represents perfect equipment effectiveness, with no downtime, optimal performance, and zero defects.[12]

Or in fair terms:

$$OEE = \frac{\text{Operating Time (h)}}{\text{Planned Time (h)}} \cdot \frac{\text{Actual Units Produced (units)}}{\text{Operating Time (h)} \cdot \text{Nominal Speed (units/h)}} \cdot \frac{\text{Total OK unit output (units)}}{\text{Actual Units Produced (units)}} = \frac{\text{Total OK unit output (units)}}{\text{Planned Time (h)} \cdot \text{Nominal Speed (units/h)}} \quad (9)$$

$$OEE = A \cdot \mu \cdot Q = \frac{\text{Total OK unit output (units)}}{\text{Planned Time (h)} \cdot \text{Nominal Speed (units/h)}} \quad (10)$$

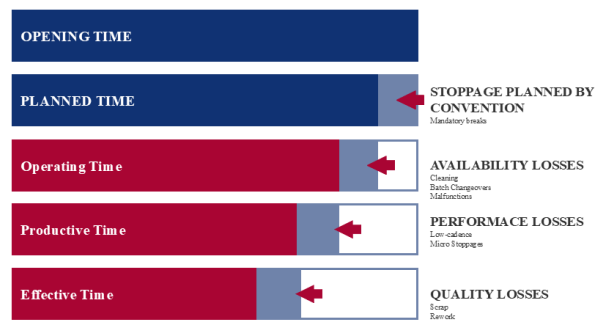


Figure 9: Visualization of OEE efficiency losses. Source: Internal Kaizen Institute documents.

2.1.2.6 Profit Calculation

Now that the concept of *OEE* is clear. In this project the focus will be on the non investment improvements. Changing the Nominal Speed of a line or machine could mean investing up to hundreds of thousands in new cutting-edge technology. Instead, in this project, the focus will be on eliminating *Muda*, and standardizing the ways of performing a task.

Creating standards and organizing the work in a scientific manner have almost no other cost than the expertise of implementing it.

The results from the project will be quantified as the augmented capacity derived from an improvement in OEE. Profits will mainly come in the form of reduced planned downtime time of the line.

In this work there will not be a focus on improving sales directly, so the Total Number of units produced will be fixed. There will be no focus on investing in new equipment, so the Nominal Speed of the line will stay constant. Every point of improvement of OEE will be inversely related to a decrement in the need for more planned time which in turns decreases the operating cost of labor of the line. Relating to the objective function depicted in the next section, equation 3.

2.1.3 Identifying improvement opportunities

A comprehensive analysis of Overall Equipment Effectiveness (OEE) provides a structured approach to identifying production inefficiencies.

Taking a closer look to the line's manufacturing flow. In Figure 10, it is clear that not all Planned time for the line is spent in production. There are gaps between one product and another where the line is down.

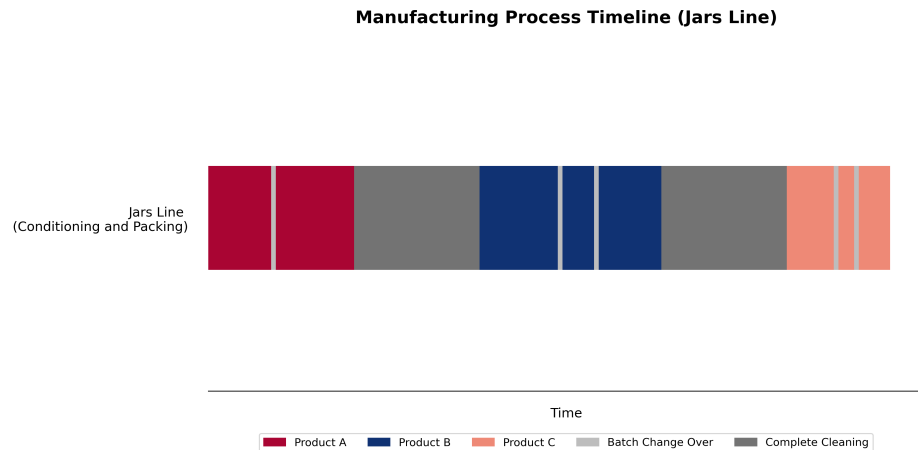


Figure 10: Manufacturing Flow for the Jars Line.

To determine how much of the actual planned time is spent in production the baseline OEE was established after one week of observation in February.

The line operators were given a production log where they would register each activity the line was performing as well as the start and end time of each activity. An example of the log used after being digitalized can be found in Appendix B Figure 29.

After the first week of observation (total of 80h planned line time) the following results were established as a baseline:

- Availability=41.6%
- Performance=66.2%
- Quality=99.89%
- OEE=27.5%

The baseline data clearly indicates that low availability is the primary factor limiting overall equipment effectiveness. This is further explained in Figure 11, which shows that only 40% of the available time is utilized for production.

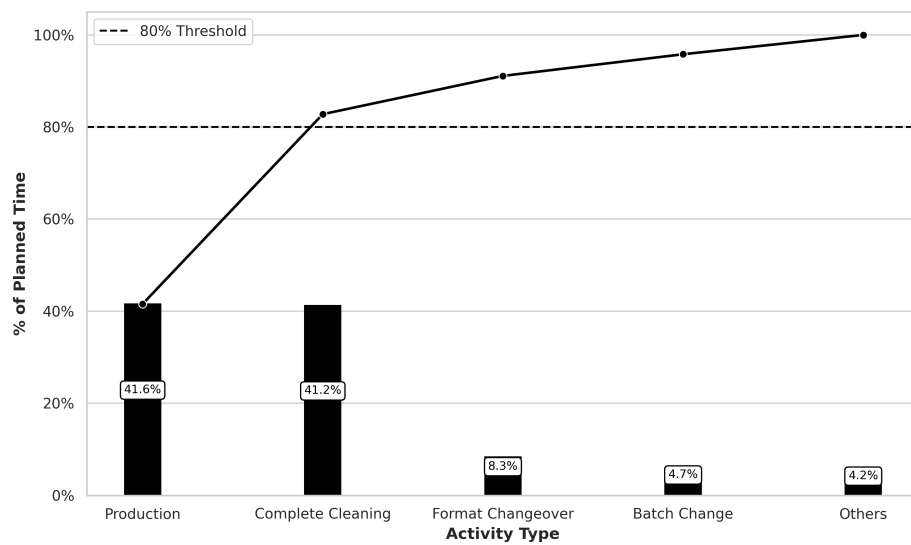


Figure 11: Pareto Chart of Planned Time dedication by Activity (%)

The main reason for this loss in availability is the time dedicated to cleaning all machine equipment and the production space. This extensive cleaning is a direct consequence of stringent quality controls within the pharmaceutical industry, which mandate:

- The prevention of cross-contamination between different products or batches.
- The removal of product from machinery overnight to prevent quality degradation.

Figure 12, provides a more detailed breakdown of the availability losses. By applying [Eighty-twenty Rule](#) the key areas for improvement can be narrowed down.

The analysis confirms that a significant efficiency improvement can be achieved by reducing the time spent on cleaning. This objective will be addressed using the [SMED](#) Methodology.

2.1.4 Implementing the SMED Methodology

2.1.4.1 SMED Overview

The potential detected in the previous section is not easy to tackle. The company have been producing the same way for over 30 years and staying regulation compliant is a must.

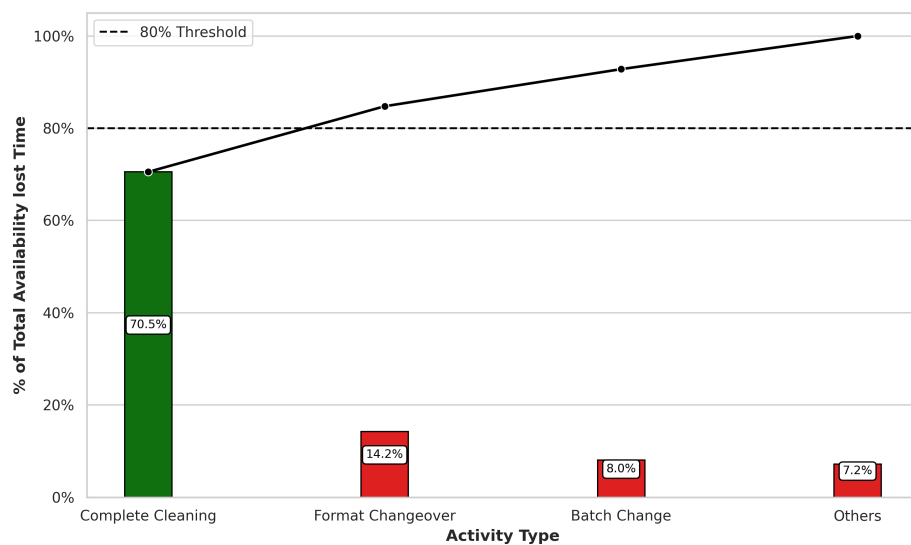


Figure 12: Pareto Chart of Main Losses on Availability. In Green Above 80% of accumulated time, in Red the rest 20% of time

The challenge radicates on applying the scientific method to resource organization. Generating an hypothesis and measuring the effect onto the project KPI.

Summarizing, in this project a detailed view on the cleaning process will be obtained, with the goals of quantifying, organizing and prioritizing cleaning tasks for a better outcome. This goal is visually exemplified on Figure 13.

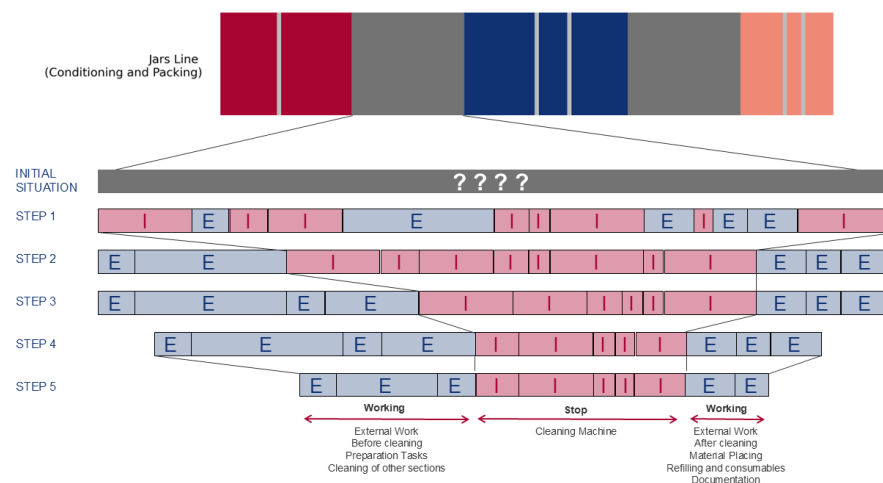


Figure 13: Detailed application overview of the SMED Methodology applied to the pilot line. Marked with a Red box are the Internal tasks. Marked with a Blue box are the External tasks.

2.1.4.2 Documentation of Tasks and Timings

The first preliminary step of the **SMED** methodology is to conduct a thorough analysis of the existing changeover process to establish a baseline. This was achieved through direct observation and detailed

time studies, utilizing video analysis to capture every task performed by the operators and the shift leader inputs to calculate their precise timings. This meticulous documentation serves to eliminate assumptions and reveal previously unnoticed process inefficiencies. A summary of the identified tasks is presented in Table 2, while the complete, detailed list can be found in the Appendix B. Note that task times are provided for every task but this are organized between 3 operators so the total cleaning time is not the addition of all tasks. Reference Appendix B for more details.

Figure 14 shows a snippet of the video recorded in order to document the process.



Figure 14: Frame of recording where the operator is performing the cleaning Task (Clean Cremer 1)

Table 2: Cleaning Tasks Sorted by Duration (Descending)

Task	Time (min)	Task	Time (min)
Clean technical area	120	Clean secondary equipment	60
Clean ceilings and walls (secondary)	45	Clean positioner	45
Clean secondary floor	30	Clean Cremer 1	30
Clean ceilings and walls	30	Remove leftover material	25
Clean T-Delta	23	Clean Cremer 2	23
Dismantle Cremer suction equipment	15	Take carts to washing area	15
Soap cart 1 parts	15	Clean floor	15
Clear production line	13	Vacuum and blow	12
Apply disinfectant cart 1	10	Dismantle positioner	10
Dismantle Cremer 1	10	Dismantle T-Delta	10
Count materials	10	Bag and label clean parts	8

2.1.4.3 Separating Internal from External Tasks

With a comprehensive understanding of the current state, the focus shifted to categorizing each task. Through collaborative workshops and direct consultation with the operators, each task was classified as either *internal* or *external*. This classification was visually represented in an *E-I* chart, which

laid the foundation for the subsequent conversion efforts. The resulting categorization is presented in Table 3.

Table 3: SMED Tasks Classified as Internal and External

Internal Task	Time (min)	External Task	Time (min)
Clean secondary equipment	60	Clean technical area	120
Clean positioner	45	Clean ceilings and walls (secondary)	45
Clean Cremer 1	30	Clean secondary floor	30
Clean ceilings and walls (primary)	30	Remove leftover material	25
Clean T-Delta	23	Clean floor	15
Clean Cremer 2	23	Vacuum and blow	12
Dismantle Cremer suction equipment	15	Count materials	10
Take carts to washing area	15	Bag and label clean parts	8
Soap cart 1 parts	15	...	...
Clear production line	13		
Apply disinfectant cart 1	10		
Dismantle positioner	10		
Dismantle Cremer 1	10		
Dismantle T-Delta	10		
...	...		
Total Internal Task Time	3h 18min	Total External Task Time	1h 47min
	Total	5h 5min	

This analysis revealed a significant opportunity for improvement. The total calculated internal time needed was a maximum of **3h 18min**, within a total cleaning time baseline of **5h 5min**. This gap represented the potential for substantial time savings. A visual representation of all operations and their timings is presented in the Appendix B, Figure 29.

2.1.4.4 Standardizing the New Procedure

To lock in the improvements and ensure their sustainability, a new standard operating procedure was developed. This was not merely a document but a comprehensive set of visual tools created in close collaboration with the operators. Every step was detailed in a clear checklist to provide a quick visual reference for the entire process.

The new procedure was treated as a living document. Any deviation from the objective time was immediately treated as a problem to be solved, then the shift leader and a small team would analyze the root cause of the delay, identify a countermeasure, and update the standard procedure to prevent the issue from reoccurring. This feedback loop ensured continuous improvement and maintained the gains achieved through the SMED effort.

The new checklist of operation added the differentiation between internal and external task, to make sure the machine availability would be hindered as less as possible from the cleaning procedure.

An example of the checklist used can be found in the Appendix B on Figure 30.

To facilitate and control the variability of some more complicated tasks, OPL were created, this standards showed exact measurement's and specific placements for the changeover. An example of a OPL in shown in Figure 15

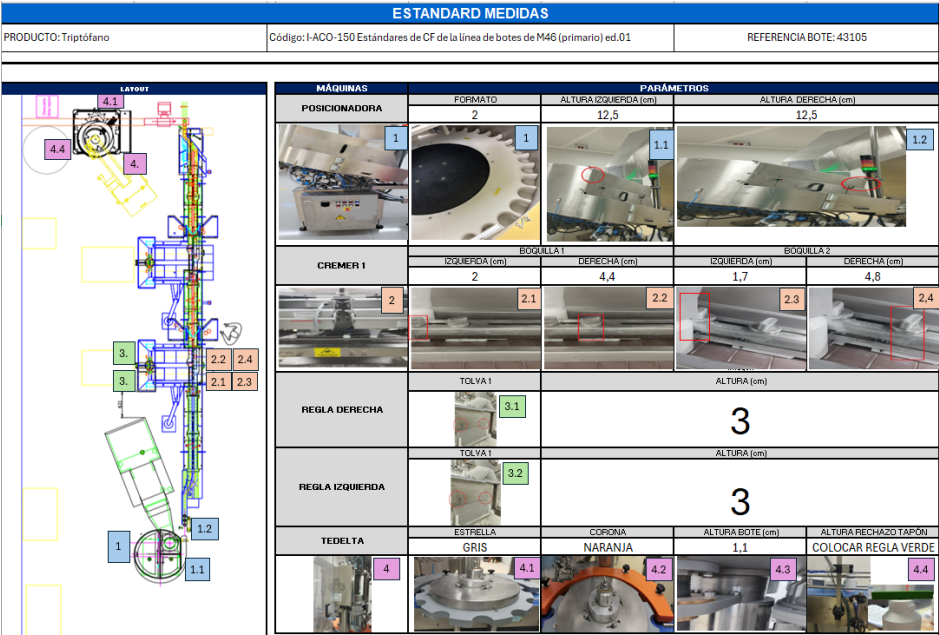


Figure 15: OPL showing measures and placement for machine parts after cleaning.

2.1.5 Results and Process Evolution

The baseline data, collected during my observations in February 2025, established the average time for the Jars line cleaning procedure at **5 hours and 5 minutes**. This total time was composed of **3 hours and 18 minutes** of internal operations (performed with the line stopped) and **1 hour and 47 minutes** of external operations that were also conducted during downtime.

Based on this analysis, a two-pronged strategic goal was established. The first and most immediate objective was to convert the **1 hour and 47 minutes** of external operations, which were previously performed with the machine off, to be conducted in parallel with production or during the preparation phase. This would immediately reduce machine downtime from 5 hours and 5 minutes to a maximum of 3 hours and 18 minutes, matching the internal operation time. The second objective, informed by the initial analysis, was to systematically reduce the remaining internal time by a third, setting an ambitious final target of **2 hours and 15 minutes** for the entire cleaning procedure.

The evolution of the cleaning process time is visually documented in Figure 16, which tracks the procedure's duration over several weeks.

The data reveals a clear and immediate reduction in downtime following the introduction of the new Standard Operating Procedure and the strategic conversion of external tasks. Machine inactivity levels dropped drastically, reflecting the direct impact of the **SMED** methodology. The figure also demonstrates a clear downward trend in cleaning time over the subsequent months, indicating a culture of continuous improvement and effective knowledge sharing among the operators. This sustained improvement is a powerful testament to the successful application of Lean principles beyond the initial project scope.

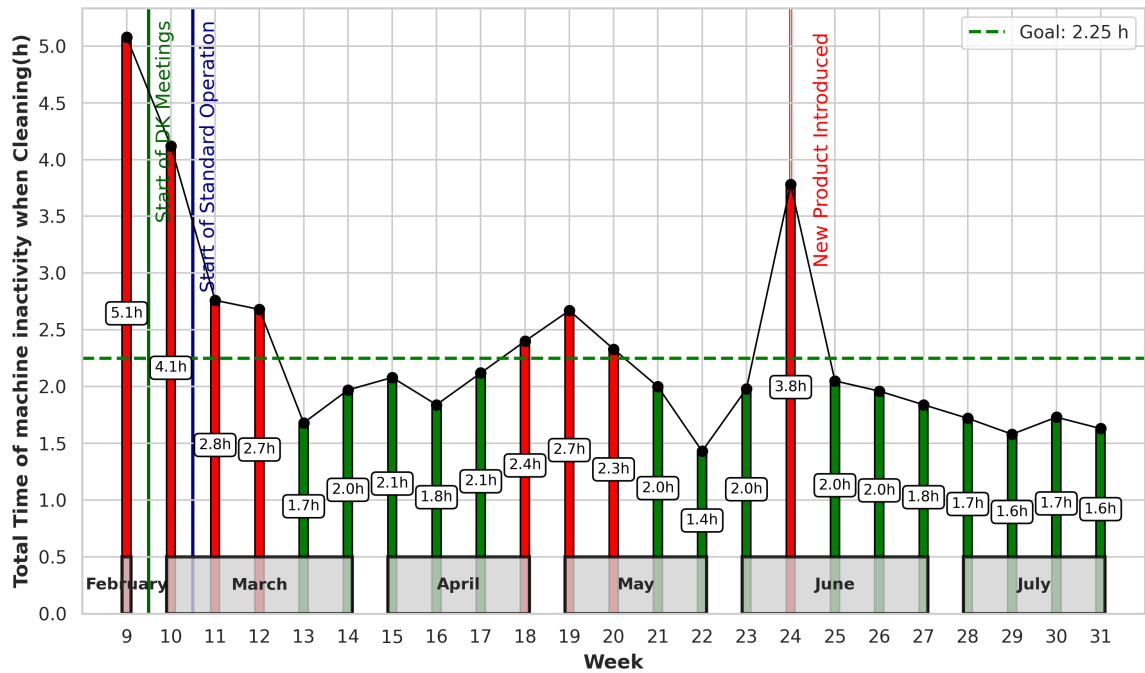


Figure 16: Evolution of the Cleaning Time Procedure along the Weeks. In red results that are above the targeted goal. In green results that are below the targeted goal.

A good example of this came in week 24, when we introduced a new, high viscosity product that disrupted the established processes. At first, the adhesive characteristics proposed an unexpected challenge threatening the steady progress on the adaptation of the new standard. However, the team quickly used the standard procedure and checklist as a guide to break the problem down and adjust on the fly. Operators worked together to test ideas, tweak tools and cleaning products. Settling on a better method. By the next week, they had updated the standard and were back on track with the time targets. This showed just how adaptable the team had become and how a strong Kaizen culture helps us solve problems quickly and lock in improvements.

As a conclusion, from the first observation of a Cleaning machine downtime of **5h 5min** to the final observation right before summer break of **1h 36min** there is an improvement of **68%** reduction on machine downtime during Cleaning procedures.

This reduction of cleaning time impacted directly to the OEE by means of the Availability metric. In the Figure 17 the evolution of the OEE within the weeks is depicted.

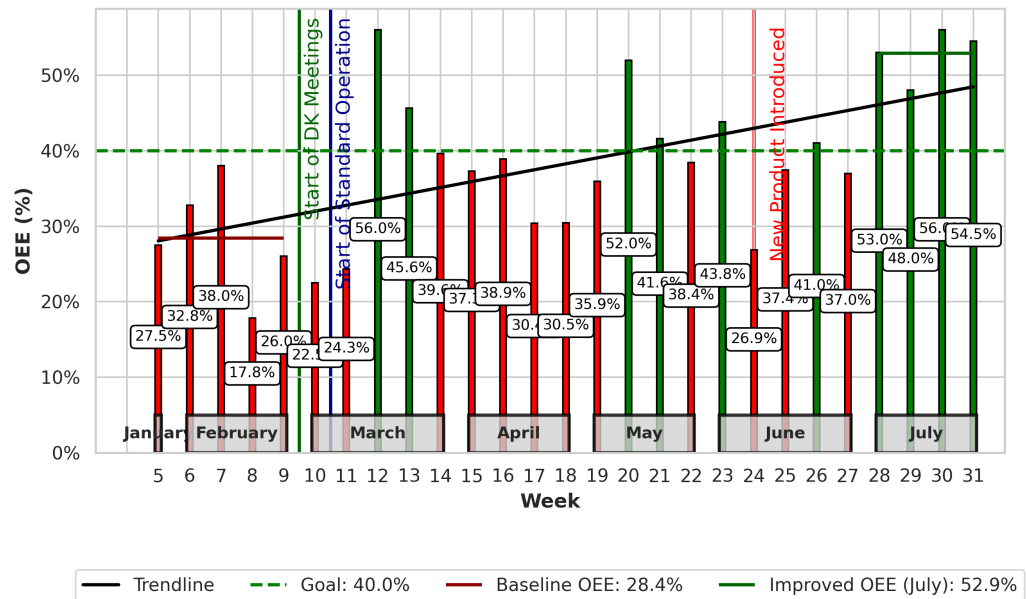


Figure 17: Evolution of OEE along the weeks

As it is shown the improved trend for the OEE is clear. From a Baseline recorded the first 5 Weeks of an average of 28.3% to an average, the last month of recorded data, of **52.9%** a 14 point improvement in OEE metrics.

2.2 Improving Line Productivity

2.2.1 Objective Function

This subproject is grounded in [GQCDM Principles](#) principles, with a primary focus on reducing direct labor costs through the optimal allocation of human resources necessary to complete a production batch. The analytical perspective shifts from traditional machine efficiency metrics to the evaluation of human time investment, identifying labor allocation as a critical lever for productivity improvement.

The objective function for this subproject is formally defined as the minimization of direct labor costs, as expressed by the following equation:

$$\begin{aligned} \text{Objective Function} &= \min (\text{Direct labor costs}) \\ &= \min \left(\text{Num. FTE assigned to line} \cdot T_{\text{operation}} (\text{h}) \cdot \text{Total labor cost (€)} \right) \quad (11) \end{aligned}$$

Where:

- *FTE*: Full Time Equivalent. Represents the total number of full time operators directly assigned to the line.
- *T_{operation}*: Denotes the total hours paid to the operator.
- Total labor cost: Refers to the labor cost for the company, including, salary, taxes, social security, etc.

Minimizing direct labor costs, as defined by this function, inherently targets production cost reduction.

2.2.2 Key Performance Indicators Selection

2.2.2.1 Productivity

In this project the KPI selected for the control of the objective function is the Productivity. This KPI directly relates to the objective function by quantifying the man time dedicated to producing a single unit.

The relation between the indicator and the objective function is visually represented in [Figure 18](#).

For this study, productivity will be calculated using the following formula:

$$\text{Productivity} = \frac{\text{Unit output (units)}}{T_{\text{operation}} (\text{h}) \cdot N_{\text{FTEs}}} \quad (12)$$

Where:

- *Unit output*: Represents the total number of units produced within a defined production batch.
- *T_{operation}*: Denotes the total person-hours invested in production, expressed as the sum of all operator working hours.
- *N_{FTEs}*: Refers to the number of Full Time Equivalent operators assigned to the production line.

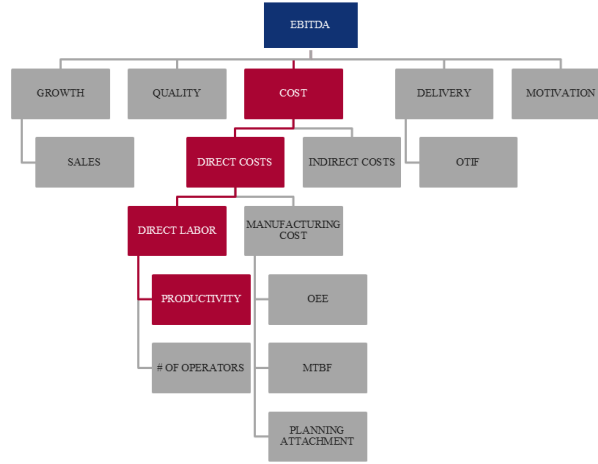


Figure 18: KPI Tree depicting relationship between selected indicators

Because products produced on the same line may exhibit different standard line speeds, this factor is incorporated to enable fair comparisons across different product types. For the remainder of the analysis, productivity improvements will be referred as Δ Productivity. Measured relative to the previous standard as described in Townsend's book *The basics of line balancing and JIT kitting* [13].

$$\Delta\text{Productivity}(\%) = \left(\frac{\left(\frac{\text{Actual Unit Output}}{\text{Planned Operation Time}_1 \cdot \text{Actual } N_{\text{operators},1}} \right)}{\left(\frac{\text{Planned Unit Output}}{\text{Planned Operation Time}_0 \cdot \text{Planned } N_{\text{operators},0}} \right)} \right) - 1 \quad (13)$$

Here, the subscript 0 refers to the baseline or historical production values, while subscript 1 corresponds to the post line-balancing measurements.

2.2.2.2 Profit Calculation

In alignment with *OEE* profit analysis, each incremental gain in productivity is translated directly into cost savings by reducing the total man time required to produce the same number of units. This approach establishes a direct quantitative link between productivity improvements and financial performance, consistent with Lean principles of continuous improvement and cost efficiency.

2.2.3 Identifying improvement opportunities

As stated in previous sections, the pilot line for this endeavour was the "Liquids" Line of the plant. In this line drugs sold and administered in liquid form are bottled, packaged, added with the dosification device necessary, boxed into distribution boxes and palletized ready for shipment.

It is divided into 2 main sections, the primary section, and the secondary. In the pharmaceutical industry, the primary section is where the product is treated outside of its shipping container, in this case its where the liquid medicine comes out from the reactor into the bottler. The primary section is regulated under heavier security measures as the product is in contact with the environment and it is more likely to be contaminated. The secondary section is where the conditioning of the bottle takes place, this is the labelling, boxing, leaflet insertion, dossier insertion, and further transformation occurs.

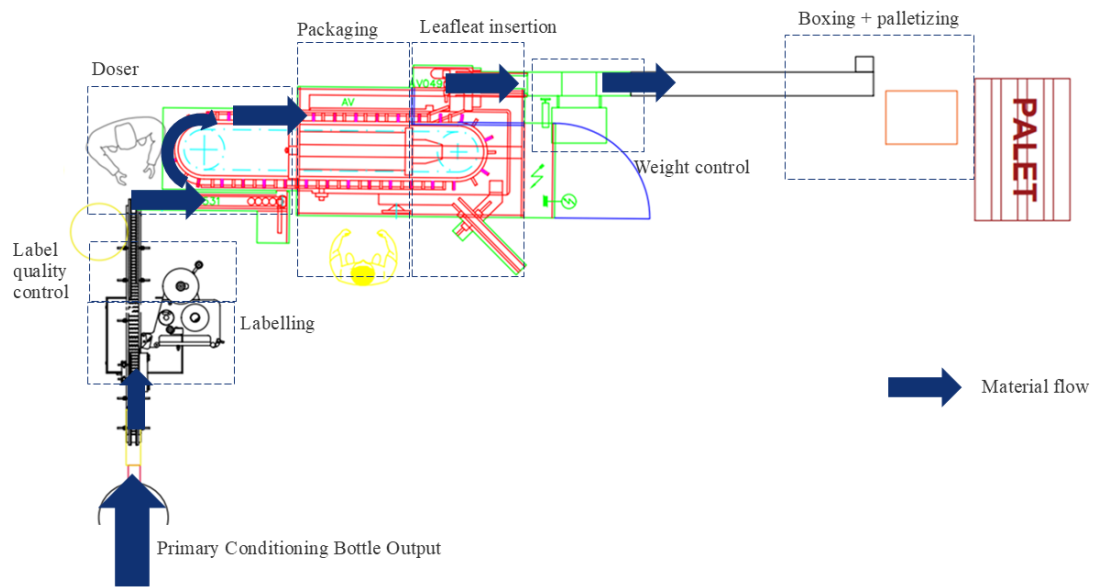


Figure 19: Detailed layout and material flow in the Liquids Line.

As the product is already bottled up and has no interaction with the environment the regulations are much looser. The two sections are physically separated by a positive pressure wall and to access the primary one must change his attire into a more restricted one which in turn difficult the movement between the two sections.

One person must reside at all times in the primary section and as movement between the two section is heavily restricted the focus of this project will be put specially on the secondary section of the line, as not only it host the most amount of resources it is where the majority of the transformation takes place.

2.2.3.1 Initial Line Distribution

As shown in Figure 20 the line hosted 2 operators and 1 line coordinator.

After an initial observation round, every task done by each operator during the shift was recorded, classified and inspected. The table showing details can be found in the Appendix C Table 7.

The line *Takt Time* was calculated to 2.8s. The Yamazumi chart in figure 21 shows task distribution between operators. It is clear that operator occupation is not maximized. This chart shows us that the need for 3 operators in the line is not due to high occupation and line tasks but rather due to a lack of time organization and task sequencing.

2.2.4 Line Balancing

The main problem was indeed time organization. Some operations had to be performed while the line was up, like putting the dosification device in the package or boxing the final product at the end of the line.

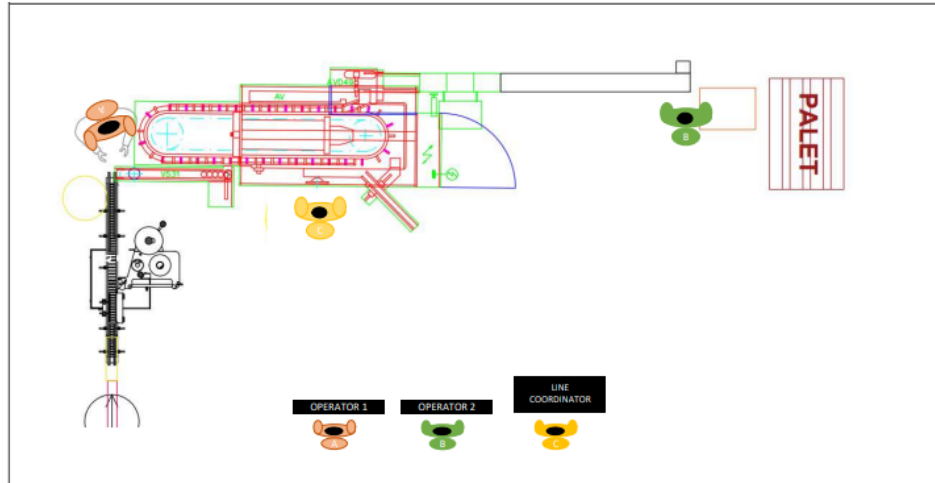


Figure 20: Line Diagram of Operator distribution

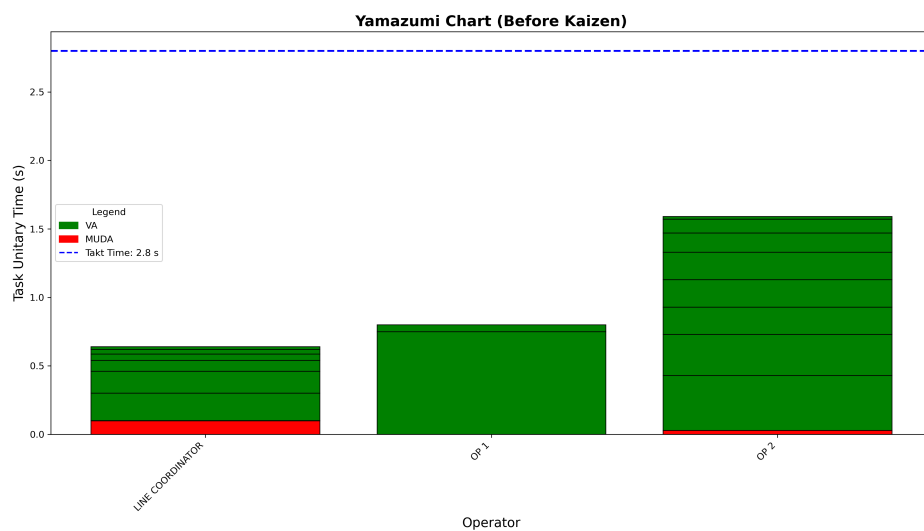


Figure 21: Yamazumi Chart Showing Task distribution before improvement.

Some other tasks, like quality checks, reloading leaflets or cases could be completed at a specific frequency. This frequencial task can be stacked up and set apart for a specific time.

For example at the current line speed, a whole loading of leaflets lasted around 5 minutes. There was no need to continuously reload the leaflet feed and this task could be reorganized.

This placed the challenge to modify the current line pace. After careful analysis it was concluded that the bottle neck for the line came from the bottling rate in the primary section. Which meant that the main secondary machine, the leaflet placer could be stopped at times and then sped up systematically in order to catch up to the primary production without ever compromising the line bottle neck. This method will be address as *Buffering* from now on, as it builds up a buffer of product on the entrance to the leaflet placer and allows operators to perform routinely tasks without compromising line efficiency.

For this purpose a buffer in the line was set up. Every 5 minutes the line would stop to perform the frequencial tasks and bottles would stack up on the buffer. Then after all the frequencial tasks

had been realized, the buffer was cleared by bumping up the line speed while maintaining this way the average line *Takt Time*.

Organizing the time in this matter and reducing the Non-added Value activities as much as possible, the line tasks where reorganized to fit only 2 operators.

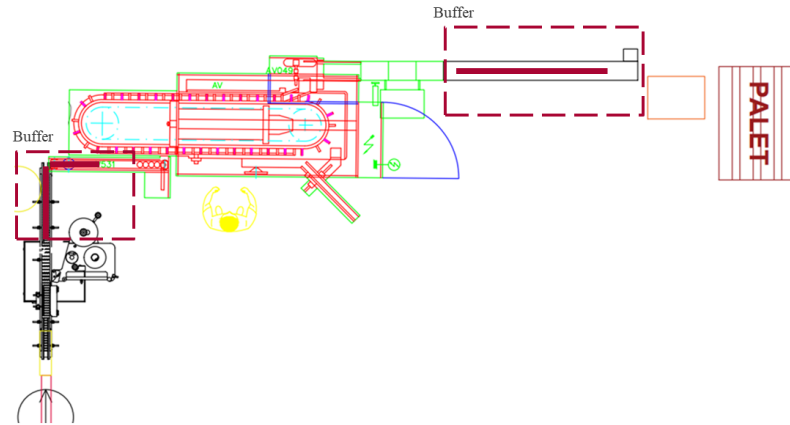


Figure 22: Liquids Lane layout showing the placement of the new buffers.

The new Yamazumi chart with only 2 operators is shown in Figure 23.

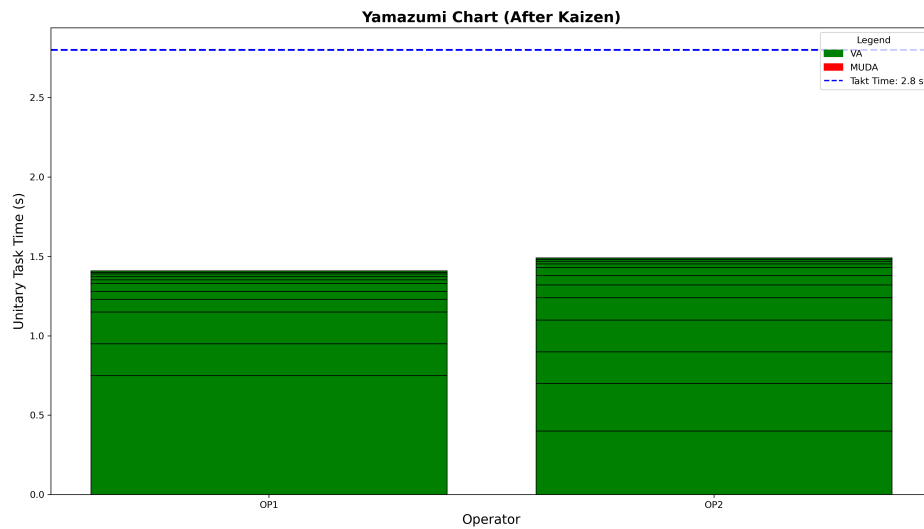


Figure 23: Yamazumi Chart Showing Task distribution with 2 operators.

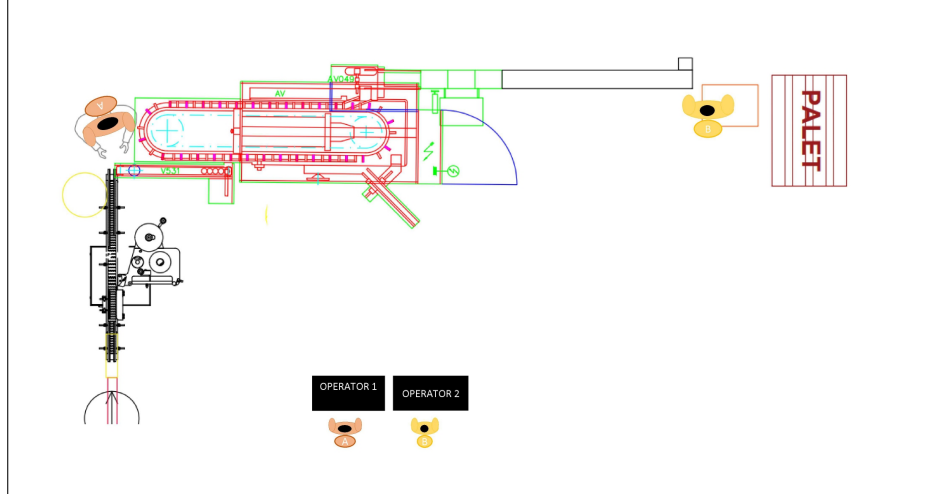


Figure 24: Line distribution of the operators with 2 operators working in the line at a time.

In the end, the deliverable of this subproject is the generation of the General Standard of the Worker, which describes how the line should be operated. What will the speed of each line machine be and how tasks will be organized with the planned resources. An example of such a standard can be found in Figure 31 on the Appendix C.

2.2.5 Results and Evaluation

Coming back to equation 13 where the method for calculating the increment in productivity was described, here, as line *Takt Time* and Unit Output stay the same, it can be simply calculated the theoretical gain on $\Delta Productivity$ as:

$$\Delta Productivity(\%) = 1 - \frac{N_{operators,1}}{N_{operators,0}} = 1 - \frac{2}{3} = 33\% \quad (14)$$

Meaning a direct increase in productivity expected should be of **33%**.

However in practice a reduced line efficiency was observed. The operators could not keep up with the tasks at hand.

In Figure 25 the evolution of the $\Delta Productivity(\%)$ can be observed. Notice that in the first week of applying the new line operator distribution, planned production was not met. As described earlier operators could not keep up with the task demand whilst maintaining line efficiency so the *Buffering* method was introduced.

This particular line only ran 9 weeks from March until June, as it is not one of the saturated lines in the plant. However for the 8 weeks that run with the new buffering method and General Standard of the Worker, productivity increased an average of **22.47%**.

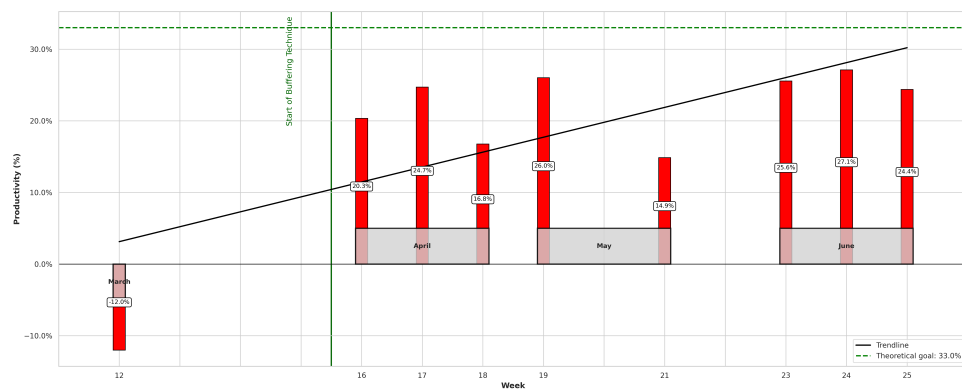


Figure 25: Evolution of $\Delta Productivity(\%)$ along the weeks

2.3 Overall Results

2.3.1 Improvement on line efficiency

In the project described on section 2.1 the main KPI for that project was **OEE**. In Figure 16 a clear downwards trend on cleaning time was observed. In fact a total reduction of over 66% of cleaning time.

2.3.1.1 Production Capacity Improvement

As explained in section 2.1.2.6 combining Equation 3 and 4 get's us the following notation:

$$\text{Production Capacity}_0 = \frac{\text{Total Output Units}}{\text{Time Window}_0} = \text{Line Rated Speed} \cdot \text{OEE}_0 \quad (15)$$

$$\text{Production Capacity}_1 = \frac{\text{Total Output Units}}{\text{Time Window}_1} = \text{Line Rated Speed} \cdot \text{OEE}_1 \quad (16)$$

As Total Output units and Line Rated Speed remain a constant and for the profit calculation method the Time Window₀ will be the entire production year, 218 days or 3,488 available hours.

$$\text{Time Window}_1 = \frac{\text{OEE}_0 \cdot \text{Time Window}_0}{\text{OEE}_1} = \frac{28.3\% \cdot 3,488h}{52.9\%} = 1,866h \quad (17)$$

Reduction of the 46% of the necessary production time was achieved, meaning that the same production output can be achieved in almost half the time.

The line currently employs 4 operators per shift, running 2 full shifts 5 days a week. By reducing the production time from 3,488 hours/year to 1,866 hours/year, saving 6,488 hours/year of direct operator working time.

The current hourly rate for a pharmaceutical operator in Spain is approximately 16.4€ before taxes [14].

A total savings of 6,488 operator hours a year represents a total savings amount of **106,400€/yearly** saved in direct labor work.

2.3.2 Improvement of line productivity

On the other hand, there is the results form the project described in section 2.2.

The Liquids Line is not a saturated line in the plant. Which mean the planned time for the line is only the 9 total weeks the exercise was run on and there weren't any other Work Orders placed in the line for the remanding of the year. The profit calculation approach follows a similar logic as the section before. The total hourly plan of the line yearly was of 720 h. From Equation 13 it is possible to isolate the parameter $\text{Operation Time} \cdot N_{\text{operators},1}$ as these are the total number of worked hours by operators.

To achieve the same production there is now a requirement to operate the line during 880 h but the total worked time by operators comes down to 1,760h versus the previous 2,160h. This savings of 400h at the same 16.4€ as previously, comes to a total saving of **6,500€/yearly**.

3 Impact Assessment

3.1 Sustainability and Ethical Analysis

The present project, responds to the industrial need for higher efficiency and competitiveness. Beyond the technical and economic outcomes, it is necessary to reflect on the environmental, social, and ethical dimensions.

3.1.1 Environmental Sustainability

Three dimensions were considered following the EEBE sustainability cell framework:

- **During the project planning and data analysis:** The main environmental impacts were related to consultancy activities (energy consumption for IT equipment, transport to the job site, and limited use of office materials). The carbon footprint was minimized through digital documentation and reduced printing.

The carbon footprint calculated for the transportation, traveling 40km per trip during 51 days. Using a gasoline car with a consumption of 7.3 l/km, the carbon footprint for transportation was of 303 kg_{CO2eq}

The carbon footprint for the IT equipment used in the 660h of a laptop consuming 65W. Taking in account the average emission rate of Spanish electricity production at 157 g_{CO2eq}/kWh [15]. The carbon footprint for IT equipment was 67,35 kg_{CO2eq} . Some measures to limit the total footprint of the project were to limit the visits to the job site to the strictly necessary.

- **During the project implementation:** The environmental impact was minimized by ensuring all chemical residues (mainly isopropyl alcohol, butanone, and other solvents) were collected and disposed of through certified hazardous waste management channels. Spill-prevention measures and proper labeling were applied, and chemical usage was optimized to avoid unnecessary waste.
- **Project Outcomes:** A full lifecycle assessment of the production output was not conducted, as it was beyond the initial scope of the project. However, the environmental impact of the implementation can be qualitatively assessed. The observed increase in production capacity and equipment efficiency directly contributes to lowering the carbon footprint per unit produced. By utilizing resources more efficiently, greater output can be achieved with the same energy input, thereby reducing the unitary environmental footprint.

3.1.2 Ethical Implications

The following ethical aspects were considered:

- **Needs addressed:** The work directly responds to the demand for reducing non-value-adding activities and increasing production efficiency. These needs were defined by the company's operational strategy but are also aligned with the broader objective of responsible resource management (contributing to several UN Sustainable Development Goals, particularly SDG 9 and SDG 12) [16]

- **Beneficiaries:** The company benefits in terms of efficiency, cost savings, and competitiveness. Employees may benefit from reduced workload pressures and more structured workflows. No collective or community has been identified as negatively affected by this work.
- **Societal benefit:** Increasing efficiency in pharmaceutical production lines contributes to more affordable medicines and greater access to essential health products, thereby generating a positive societal impact.
- **Unintended consequences:** Potential risks include excessive focus on labor cost savings that could be misinterpreted as justification for workforce reductions. The project therefore emphasizes the use of time savings to reinvest in higher-value tasks and continuous improvement activities.
- **Deontological aspects:** The work respects professional ethical standards, ensuring transparency, data integrity, and alignment with industrial safety and regulatory frameworks.



Figure 26: SDG (Sustainable Development Goals) by United Nations [16].

From both ethical and sustainability perspectives, the project is aligned with the principles of responsible engineering practice. It fosters efficiency without compromising workers' rights, improves resource utilization, and contributes positively to the environmental performance of pharmaceutical operations. In this way, it supports the integration of Kaizen methodologies not only as a tool for economic gain but also as a driver of sustainable industrial development.

3.2 Economic Evaluation

3.2.1 Project Costs

After presenting the results the topic of total project cost should be addressed. All calculations for consultancy cost will be quoted using the Spanish statute for Business Consultancy regulated statute group A3GAN1 [17], and adding to it the social charge of the 38%. Which states an hourly wage of 23€/h.

As this work was done under a context of consultancy, an extra 30% commercial margin will be applied to all project costs in order to cover for the consultancy enterprise organization and profit.

The cost evaluation will take into account:

- Direct Costs:

Total Consultancy hours: 660h

Transportation Costs: 12€ per day worked at job site.

Software Subscription (Office 365 Package): 657€

Office Materials (Paper, pens, sticky notes...): 500€ fixed cost.

- Indirect Costs

Consultant Laptop Amortization over 5 year lifetime: 19.98€/month

- Strategic Costs:

Commercial Margin: 30% of all cost

Cost Category	Basis / Rate	Amount [€]
Direct Costs		
Consultancy Hours	660h · 23 €/h	15,180
Transportation	12 €/day · (51 days)	612
Software Subscription (Office 365)	Fixed annual license	657
Office Materials	Fixed	500
Subtotal Direct Costs		16,949
Indirect Costs		
Laptop Amortization	19.98 €/month × (6 months)	119.88
Subtotal Indirect Costs		119.88
Strategic Costs		
Comercial Margin	30% of (Direct + Indirect)	5,120
Total Project Cost		22,189 €

Table 4: Cost evaluation for the project

3.2.2 Project Benefits

As advanced in section 2.3 the project calculated profits would be:

SubProject	KPI	Baseline	Endpoint	Benefit (€)
Line Efficiency Improvement	OEE	27.5%	52.9%	106,400 €/ yearly
Line Productivity Improvement	Δ Productivity	0%	22.47%	6,500 €/yearly
Total				112,900 €/ yearly

Table 5: KPI Improvements and Benefits

3.2.3 Return on Investment

Overall the project annual ROI is calculated as:

$$ROI(\%) = \frac{112,900 - 22,189}{22,189} = 410\% \quad (18)$$

Meaning the full project cost will be payed off in just **3 months** after the full implementation.

4 Conclusions

The projects described demonstrate the tangible benefits of applying Lean manufacturing tools, in particular **SMED** and line balancing techniques, to pharmaceutical production lines. Both approaches were implemented with the goal of reducing non-value-added time, increasing efficiency, and improving the utilization of available resources.

The **SMED** initiative led to a significant reduction in cleaning and changeover times, achieving a decrease of over 66%. This improvement directly impacted the line's *OEE*, which rose from a baseline of 28.3% to 52.9%, representing an increase of 24.6 percentage points. Exceeding the original goal by almost 13 points.

The resulting gain in production capacity translated into a reduction of production hours from 3,488 to 1,866 per year. When expressed in labor costs, this corresponds to savings of approximately **106,400€ per year**.

In parallel, the line balancing project demonstrated that productivity improvements can be achieved even in non-saturated lines. By redistributing operator tasks and improving workflow, the total operator working hours required for equivalent output decreased by 400 hours annually. Productivity was increased by 22.5% just 10 percentage points below the theoretical limit, and almost doubling the initial goal of a 12.5% boost in productivity.

Although the direct monetary savings (**6,500€ per year**) were modest, the initiative proved the concept's potential and established a foundation for deployment in higher-demand lines.

Together, these projects generated total benefits of **112,900€**, achieving a payback period of only **3 months**. Beyond the economic impact, they reinforced the value of lean methodologies in a highly regulated pharmaceutical environment, improving both technical performance indicators and organizational culture. The initiatives also enhanced operator engagement and fostered a mindset of continuous improvement, strengthening the company's operational resilience.

Future work could extend these approaches to other production lines and integrate digital tools for real-time **KPI** monitoring. With the company now experienced in applying the scientific method as part of its Production System, such efforts will be easier to scale and sustain, driving further improvements in efficiency and competitiveness.

5 References

- [1] James P. Womack. *The Machine that changed the world: how Japan's secret weapon in the global auto wars will revolutionize western industry*. In collab. with Daniel T. Jones and Daniel Roos. New York: HarperPerennial, 1991. viii+323. ISBN: 978-0-06-097417-6.
- [2] Taiichi Ohno. *Toyota Production System: Beyond Large-Scale Production*. CRC Press, Mar. 1, 1988. 182 pp. ISBN: 978-0-915299-14-0.
- [3] Horace Bookwalter Drury. "Chapter I. The Meaning of Scientific Management". In: *Scientific Management: A History and Criticism*. Columbia University Press, May 6, 2019, pp. 15–30. ISBN: 978-0-231-89022-9. URL: <https://www.degruyterbrill.com/document/doi/10.7312/book92740-002/html> (visited on 09/25/2025).
- [4] Imai Massaki and Cristina Sanchez. *Gemba Kaizen. Un enfoque hacia la mejora continua de la estrategia*, 2^aEd. Sept. 15, 2015. 580 pp.
- [5] MASA AKI IMAI. *Strategic KAIZEN: Using Flow, Synchronization, and Leveling [FSL] Assessment to Measure and Strengthen Operational Performance*. 1st Edition. McGraw-Hill Education, 2021. ISBN: 978-1-260-14383-6. URL: <https://www.accessengineeringlibrary.com/content/book/9781260143836> (visited on 09/02/2025).
- [6] Gajanan Gambhire et al. "Productivity Improvement in Refrigerator Manufacturing Plant by Using Yamazumi—Line Balancing Technique". In: *Optimization of Production and Industrial Systems*. Ed. by Arvind Bhardwaj, Pulak Mohan Pandey, and Aviral Misra. Singapore: Springer Nature, 2024, pp. 173–182. ISBN: 978-981-99-8343-8. DOI: [10.1007/978-981-99-8343-8_15](https://doi.org/10.1007/978-981-99-8343-8_15).
- [7] Torbjorn H. Netland and Daryl J. Powell, eds. *The Routledge Companion to Lean Management*. New York: Routledge, Dec. 8, 2016. 500 pp. ISBN: 978-1-315-68689-9. DOI: [10.4324/9781315686899](https://doi.org/10.4324/9781315686899).
- [8] Jeffrey K. Liker and David Meier. *Toyota Way Fieldbook*. 1st Edition. McGraw-Hill Education, 2006. ISBN: 978-0-07-144893-2. URL: <https://www.accessengineeringlibrary.com/content/book/9780071448932> (visited on 08/19/2025).
- [9] Mark Graban. *Lean Hospitals: Improving Quality, Patient Safety, and Employee Satisfaction*. Google-Books-ID: r92d9xVvsXkC. CRC Press, Mar. 23, 2011. 278 pp. ISBN: 978-1-4200-8499-3.
- [10] Andrew P. Dillon and Shigeo Shingo. *A Revolution in Manufacturing: The SMED System*. Google-Books-ID: ooXVVIqEQwC. CRC Press, Apr. 1, 1985. 394 pp. ISBN: 978-0-915299-03-4.
- [11] Paulo Roberto Calandrelli, Pablo Deivid Valle, and Fernando Deschamps. "Maximizing operational efficiency with Industry 4.0 technology: integrating OEE as a performance indicator". In: *The International Journal of Advanced Manufacturing Technology* 138.3 (May 1, 2025), pp. 855–872. ISSN: 1433-3015. DOI: [10.1007/s00170-025-15408-y](https://doi.org/10.1007/s00170-025-15408-y). URL: <https://doi.org/10.1007/s00170-025-15408-y> (visited on 08/21/2025).
- [12] Edward S. Pound. *Factory physics for managers: how leaders improve performance in a post-lean six sigma world*. In collab. with Mark L. Spearman and Jeffrey H. Bell. 1st edition. McGraw-Hill's AccessEngineering. New York: McGraw-Hill Education, 2014. ISBN: 978-0-07-182250-3.
- [13] Beverly Townsend. *The basics of line balancing and JIT kitting*. Boca Raton, FL: CRC Press, 2012. xi+73. ISBN: 978-1-4398-8237-5.



- [14] CCOO. *Convenio Colectivo Estatal para el Comercio de Distribuidores de Especialidades y Productos Farmacéuticos*. URL: https://industria.ccoo.es/Distribuidores_de_productos_farmaceuticos/Convenio (visited on 09/15/2025).
- [15] ElectricityMaps. *Spain / App / Electricity Maps*. Electricity Maps. URL: <https://app.electricitymaps.com/map/zone/ES/72h/hourly> (visited on 09/16/2025).
- [16] UN. *THE 17 GOALS / Sustainable Development*. United Nations. URL: <https://sdgs.un.org/es/goals> (visited on 09/16/2025).
- [17] Ministerio de Trabajo y Economía Social. *Resolución de 4 de abril de 2025, de la Dirección General de Trabajo, por la que se registra y publica el XIX Convenio colectivo estatal de empresas de consultoría, tecnologías de la información y estudios de mercado y de la opinión pública*. Apr. 16, 2025. URL: [https://www.boe.es/eli/es/res/2025/04/04/\(10\)](https://www.boe.es/eli/es/res/2025/04/04/(10)) (visited on 09/16/2025).

Appendices

A Project Details

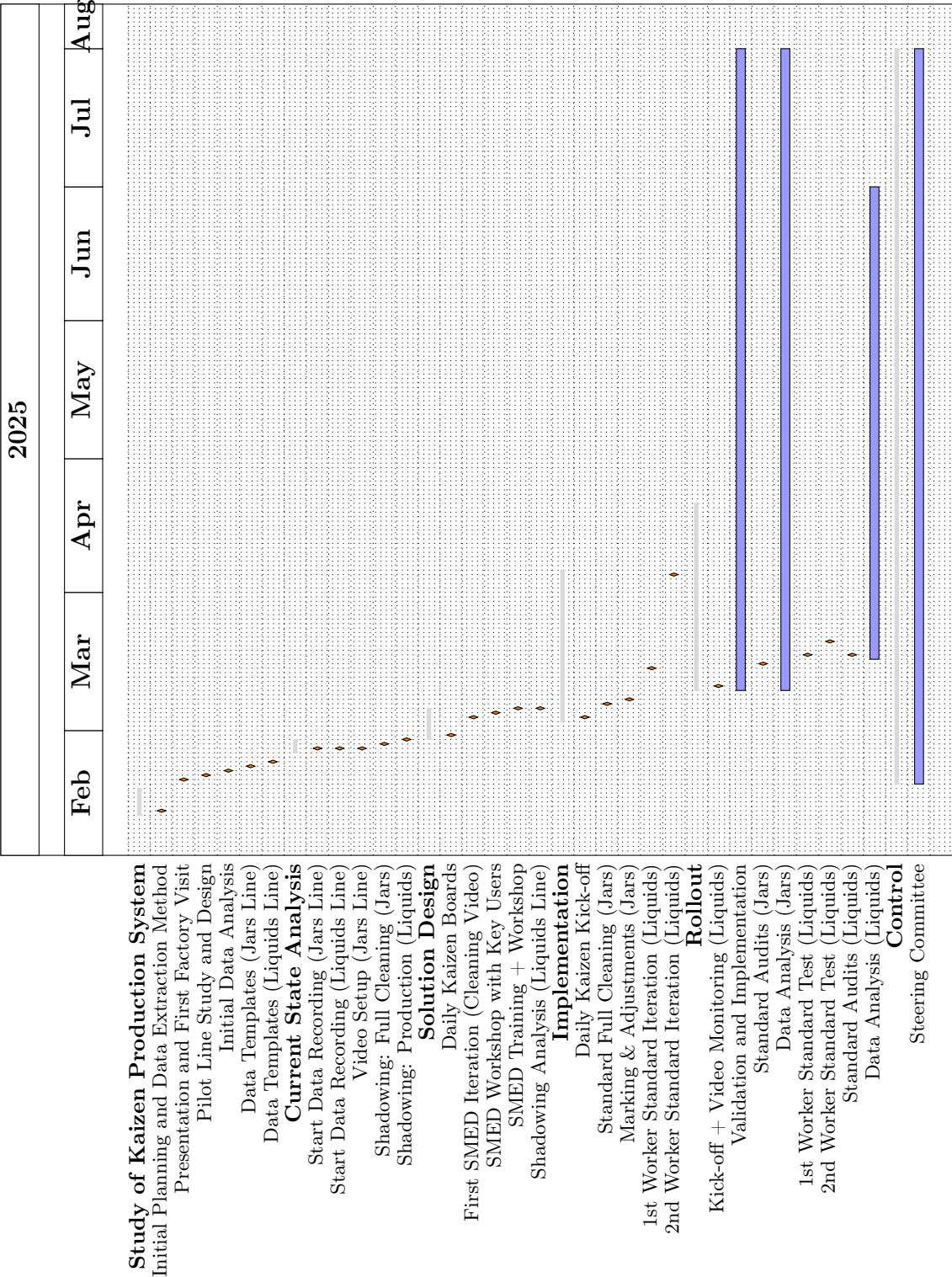


Figure 27: Detailed Project Gantt chart

B SMED Task Detail

Dia	Hora inicio	Hora fin	P/LG/CF/CL	Cantidad producida	Cantidad teórica	Rendimiento	Observaciones	Tempo no	Comprob
17/03/2025	6:45	13:45	P		12600	0			
17/03/2025	14:15	17:20	P		5550	0			
17/03/2025	17:20	17:30	CL					0:10	
17/03/2025	17:30	22:00	P		8100	0			
18/03/2025	6:15	9:30	P		5850	0			
18/03/2025	9:30	10:00	CL					0:30	
18/03/2025	10:00	14:00	P		7200	0			
18/03/2025	14:15	16:50	P		4650	0			
18/03/2025	16:50	17:50	CL					1:00	
18/03/2025	17:50	22:00	P		7500	0			
19/03/2025	6:20	13:30	P		12900	0			
19/03/2025	14:15	19:15	P		9000	0			
19/03/2025	19:15	19:35	CL					0:20	
19/03/2025	19:35	22:00	P		4350	0			
20/03/2025	6:30	13:40	P		12900	0			
20/03/2025	14:15	19:48	P		9990	0			
20/03/2025	19:50	22:00	LC					2:10	
21/03/2025	6:15	11:50	LC					5:35	
21/03/2025	11:50	14:00	CF					2:10	
21/03/2025	14:30	18:30	CF					4:00	
21/03/2025	19:45	22:00	P		4050	0			
24/03/2025	6:25	13:30	P		12750	0			
24/03/2025	14:15	16:30	P		4050	0			
24/03/2025	17:00	22:00	LC					5:00	
25/03/2025	6:00	7:20	LC					1:20	
25/03/2025	7:30	10:00	CF					2:30	
25/03/2025	11:35	13:30	P		3450	0			
25/03/2025	14:45	18:30	P		6750	0			
25/03/2025	19:00	22:00	LC					3:00	
26/03/2025	6:15	8:50	CF					2:35	
26/03/2025	9:00	13:30	P		8100	0			
26/03/2025	14:45	21:30	P		12150	0			
27/03/2025	6:25	13:30	P		12750	0			
27/03/2025	14:20	21:30	P		12900	0			
28/03/2025	6:15	6:50	P		1050	0			

Figure 28: Example of Production Log

Table 6: Cleaning and Production Tasks Sorted by Duration (Descending)

Task	Time (min)	Task	Time (min)
Clean technical area	120	Clean secondary equipment	60
Clean ceilings and walls (secondary)	45	Clean positioner	45
Clean secondary floor	30	Clean Cremer 1	30
Clean ceilings and walls	30	Remove leftover material	25
Clean T-Delta	23	Clean Cremer 2	23
Dismantle Cremer suction equipment	15	Take carts to washing area	15
Soap cart 1 parts	15	Clean floor	15
Clear production line	13	Vacuum and blow	12
Apply disinfectant cart 1	10	Dismantle positioner	10
Dismantle Cremer 1	10	Dismantle T-Delta	10
Count materials	10	Bag and label clean parts	8
Final rinse cart 1	8	Remove cleaning materials	8
Partial soaping cage 1	8	Bring cleaning materials upstairs	5
Bring parts upstairs	5	Close logs and cards	5
Access primary area	5	Initial rinse cart 1	3
Initial rinse cage 1	3	Remove materials	3
Bring carts	3	Close guide	3
Record in production log	3	Change cards	3
Remove leftover material (conditioned)	3	Bring vacuum cleaner	3
Take cart 1 to dryer	1	Insert cart 1 into washer	1

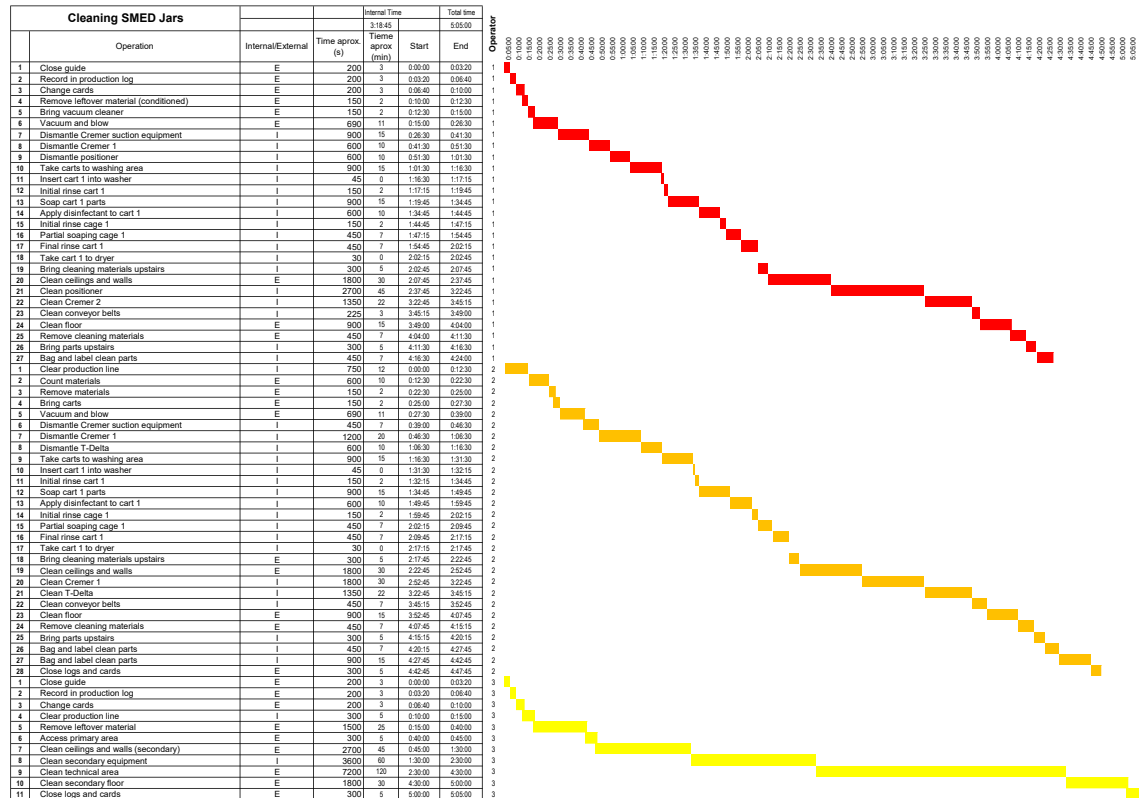


Figure 29: Gantt Diagram of Cleaning Procedure

CHECK-LIST DE LIMPIEZA					
Linea: BOTES PRIMARIO				Turno: _____	
OPERARIO 1				Fecha: _____	
Nº	TAREA	NOMBRE OPERARIO/A	HORA INICIO	HORA FIN	TIEMPO REAL
1	CERRAR GUIA				3
2	ANOTAR EN DIARIO DE PRODUCCIÓN				3
3	TARJETAS CAMBIO COLOR				3
4	SACAR MATERIAL SOBRANTE(ACO)				3
5	TRAER ASPIRADORES,Y ASPIRAR(2 PERS)				23
6	DESMONTAR EQUIPO SUCCION CREMER (2PERS)				15
7	DESMONTAR CREMER 2 (OPCIONAL)				
8	DESMONTAR POSICIONADORA				10
9	SOPLAR				10
10	BAJAR CARROS, JAULAS AL LAVADERO				15
11	LIMPIEZA PIEZAS LAVADERO (2PERS)				5
12	PASAR AGUA PIEZAS				5
13	PASAR JABON PIEZAS				30
14	ACLARAR PIEZAS				15
15	ECHAR DESINFECTANTE (20MIN)				20
16	ACLARAR PIEZAS (3MIN)				5
17	DEJAR PIEZAS EN EL SECADERO				5
18	SUBIR MATERIAL LIMPIEZA				15
19	LIMPIEZA TECHO Y PAREDES(2PERS)				30
20	LIMPIEZA POSICIONADORA				45
21	LIMPIEZA CREMER 2 (OPCIONAL)				
22	LIMPIEZA PIEZAS TRANSPORTADORA				15
23	LIMPIEZA ESCALERAS MESA MATERIAL ADICIONAL				7
24	LIMPIAR SUELO(2 PERS)				30
25	BAJAR MATERIAL DE LIMPIEZA AL LAVADERO				5
26	SUBIR PIEZAS SECAS(EMBOLSAR Y ETIQUETAR PIEZAS LIMPIAS)(2 PERS)				15

Figure 30: Standard Procedure Checklist Example



C Line Balancing Details

Tasks	Time (s)	Frequency (1/s)	MUDA/VA	Operator
Change dispenser box	50	0.001	VA	OP 1
Change case box	50	0.0004	VA	OP 2
Change leaflet box	50	0.0004	VA	LINE COORDINA- TOR
Change label	70	0.0005	VA	LINE COORDINA- TOR
Load cases	20	0.004	VA	LINE COORDINA- TOR
Load leaflets	40	0.004	VA	LINE COORDINA- TOR
Check machine parameters	10	0.0002	MUDA	LINE COORDINA- TOR
Case quality control	20	0.0022	VA	LINE COORDINA- TOR
Record quality control result	45	0.0022	MUDA	LINE COORDINA- TOR
Monitor rejection 2	10	0.02	VA	OP 2
Discard boxes	84	0.0012	VA	OP 2
Perform label rejection control (Rejection 1)	20	0.01	VA	LINE COORDINA- TOR
Pallet shipment	180	0.00078	VA	OP 2
Push bottles onto the car- toner conveyor	30	0.001	MUDA	OP 2
Place dispenser in case	0.75	1	VA	OP 1
Label box	1.5	0.2	VA	OP 2
Place cases into boxes (5 at a time)	1	0.2	VA	OP 2
Palletize	1	0.2	VA	OP 2
Make box	2	0.2	VA	OP 2

Table 7: Line Tasks Before Improvement

ESTANDAR GENERAL DE TRABAJO				Código:			
				Fecha:	06/03/2025		
Denominación Operación:			Denominación Proceso:				
SECUNDARIO			ENVASADO				
Referencia	Denominación de la referencia	Área de operación	Ritmo uds./hora	Equipos protección individuales:			
		SALA DE ENVASADO	1800				
			Rotación	Estándar 2 personas Cada 2 horas: AxB, BxA			
OPERACIONES Y BALANCEO							
OPERACION POR PUESTOS DE TRABAJO							
OPERARIO A: PUESTO 1 Estuchado de producto		OPERARIO B: PUESTO 2 Estuchado de producto		VELOCIDAD ESTUCHADORA (uds/min)			
1. Meter dosificador en estuche 2. Cargar prospectos 3. Cargar estuches 4. Control calidad estuche 5. Cambiar caja de dosificadores 6. Controlar el rechazo 1 7. Empujar botes a cinta de la estuchadora 8. Cambiar etiqueta		1. Meter estuches 5 en 5 cajas 2. Paletizar 3. Cargar estuches 4. Cargar prospectos 5. Cambiar caja de estuches 6. Empujar botes a cinta de la estuchadora 7. Controlar el rechazo 2 8. Expedición de pallet		33			
Ritmo de trabajo: 1.800 uds./h		Ritmo de trabajo: 1.800 uds./h					
VISADOS	<CARGO>	<CARGO>	<CARGO>	<CARGO>	Observaciones	Fecha:	Revisiones
						06/03/2025	Creación del documento
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Figure 31: General Standard of the Worker Standard